Digital Image Processing

Image Enhancement:
Filtering in the Frequency Domain

Jean Baptiste Joseph Fourier



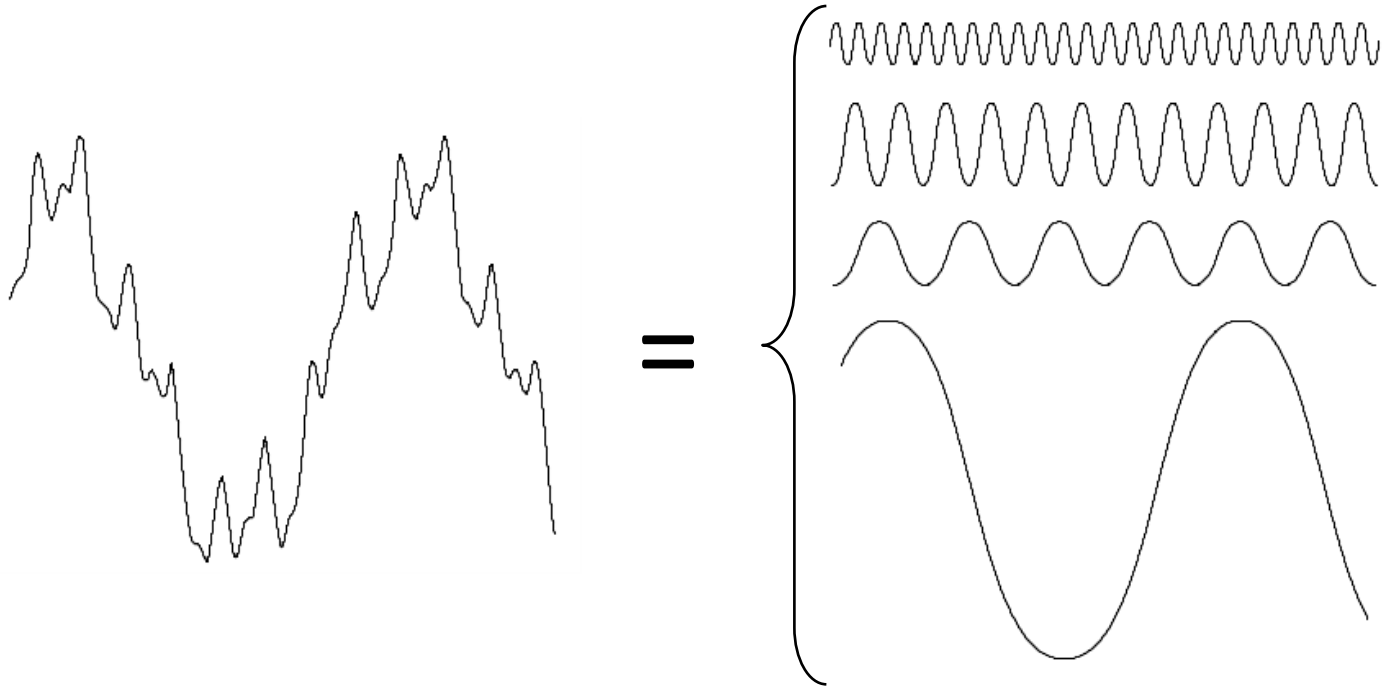
Fourier was born in Auxerre, France in 1768

- Most famous for his work "*La Théorie Analitique de la Chaleur*" published in 1822
- Translated into English in 1878: "*The Analytic Theory of Heat*"

Nobody paid much attention when the work was first published

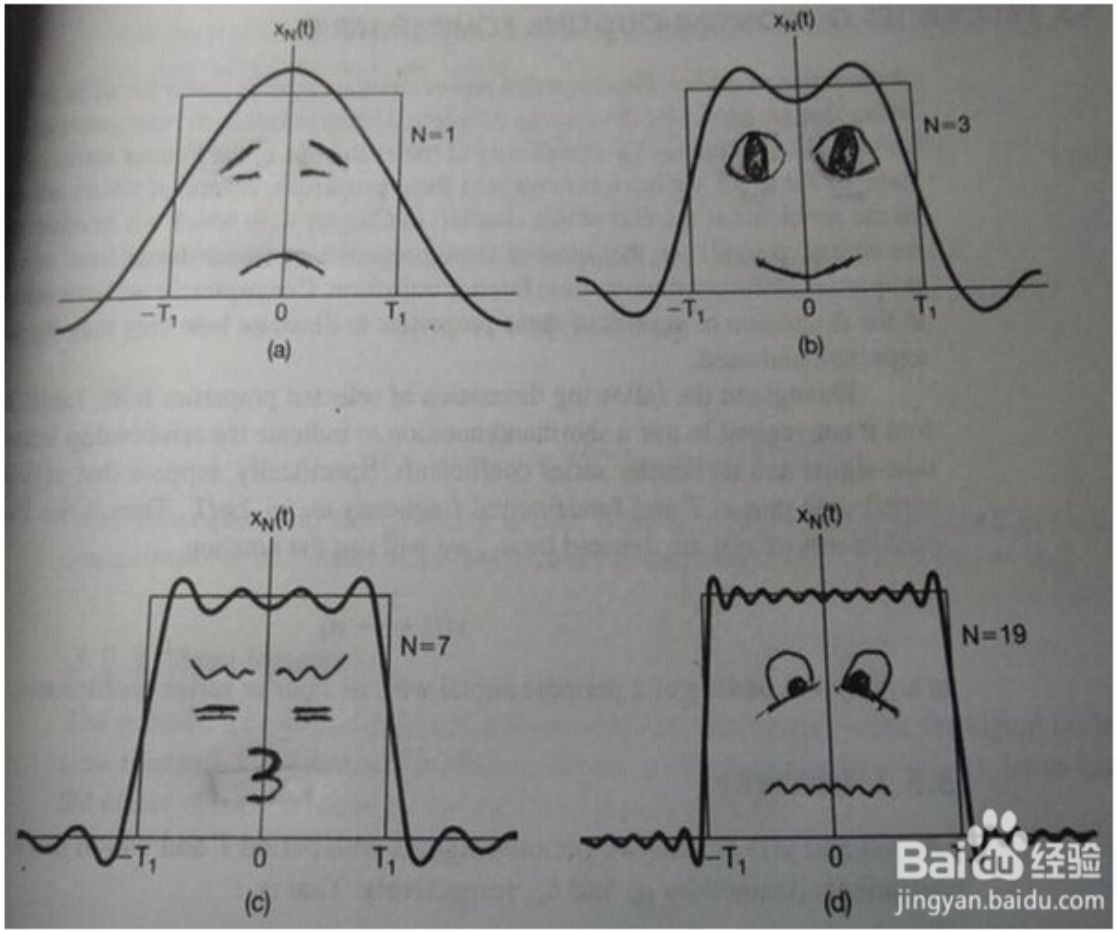
One of the most important mathematical theories in modern engineering

The Big Idea



Any function that periodically repeats itself can be expressed as a sum of sines and cosines of different frequencies each multiplied by a different coefficient – a *Fourier series*

Images taken from Gonzalez & Woods, Digital Image Processing (2002)



Baidu 经验
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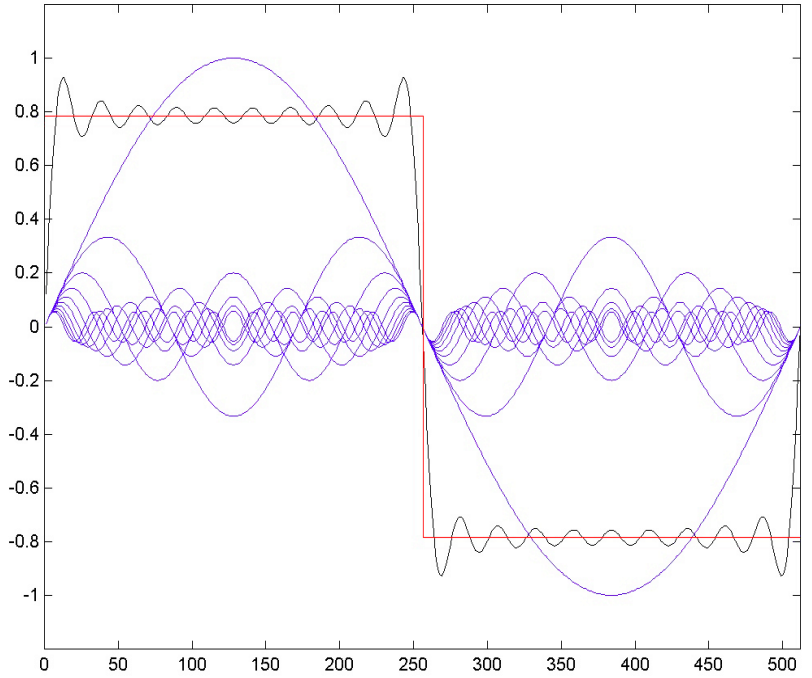
The Big Idea (cont…)

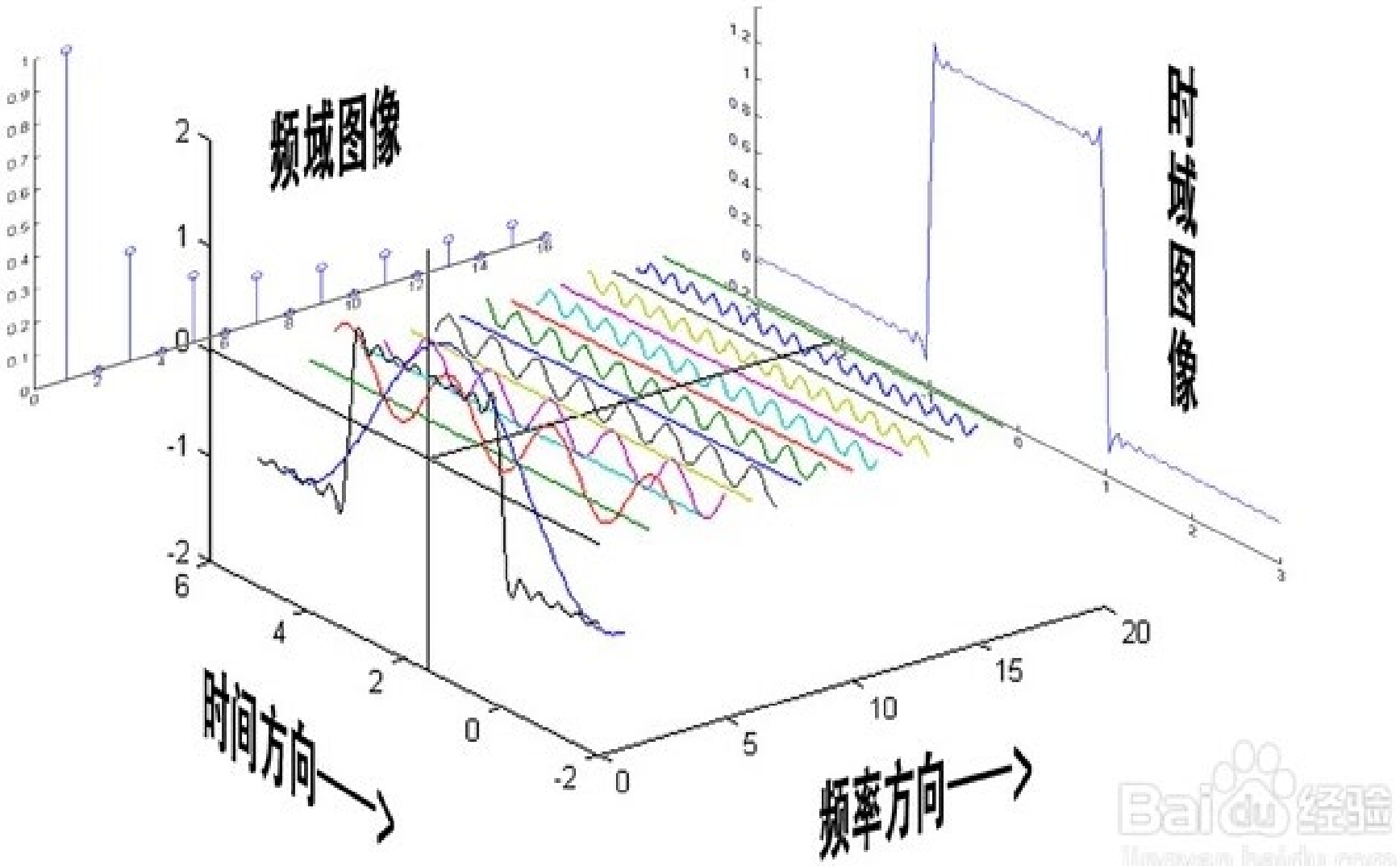
Any periodic signal can be described by a sum of sinusoids.

$$\mathrm{sq}(t) = \sum_{n=-\infty}^{\infty} \frac{1}{2n+1} \sin\left[\frac{2\pi}{\lambda}(2n+1)t\right]$$

The sinusoids are called “basis functions”.

The multipliers are called “Fourier coefficients”.




频域图像
时域图像
时间方向
频率方向
2
1
0
-1
-2
6
4
2
0
-2
0
5
10
15
20

4.5.5 The Discrete Fourier Transform (DFT)

The *Discrete Fourier Transform* of $f(x, y)$, for x = 0, 1, 2…M-1 and y = 0,1,2…N-1, denoted by $F(u, v)$, is given by the equation:

$$F(u,v) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(ux/M+vy/N)}$$

for u = 0, 1, 2…M-1 and v = 0, 1, 2…N-1.

$$f(x,y) = \frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)e^{j2\pi(ux/M+vy/N)}$$

DFT & Images

The DFT of a two dimensional image can be visualised by showing the spectrum of the images component frequencies

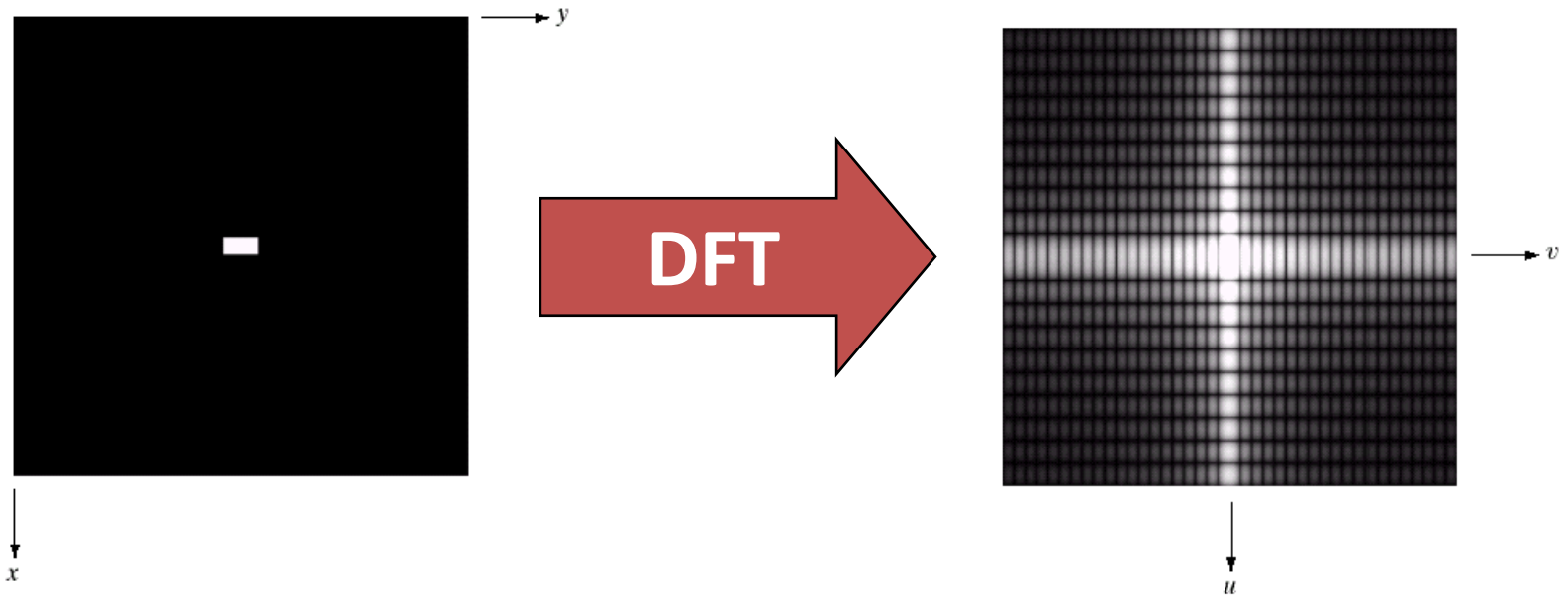


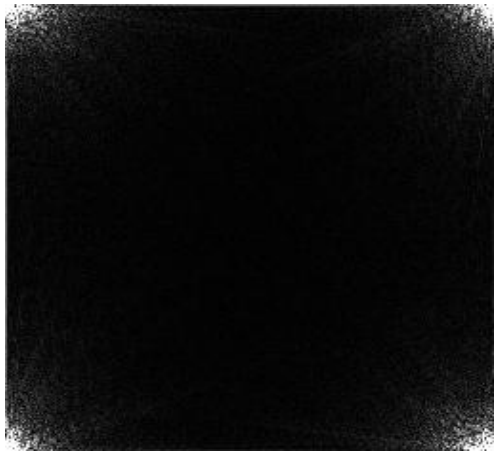
Fig 4.24

F(u,v) = F(-u,-v)

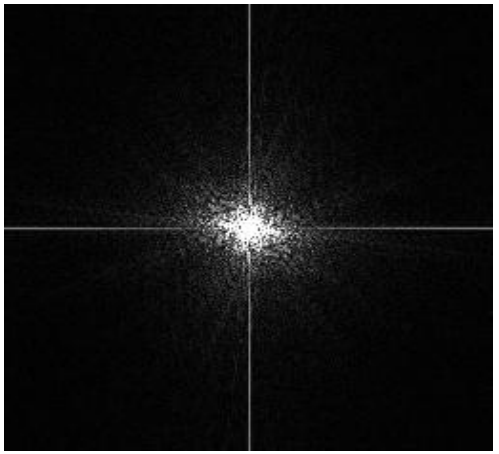
Images taken from Gonzalez & Woods, Digital Image Processing (2002)



原图

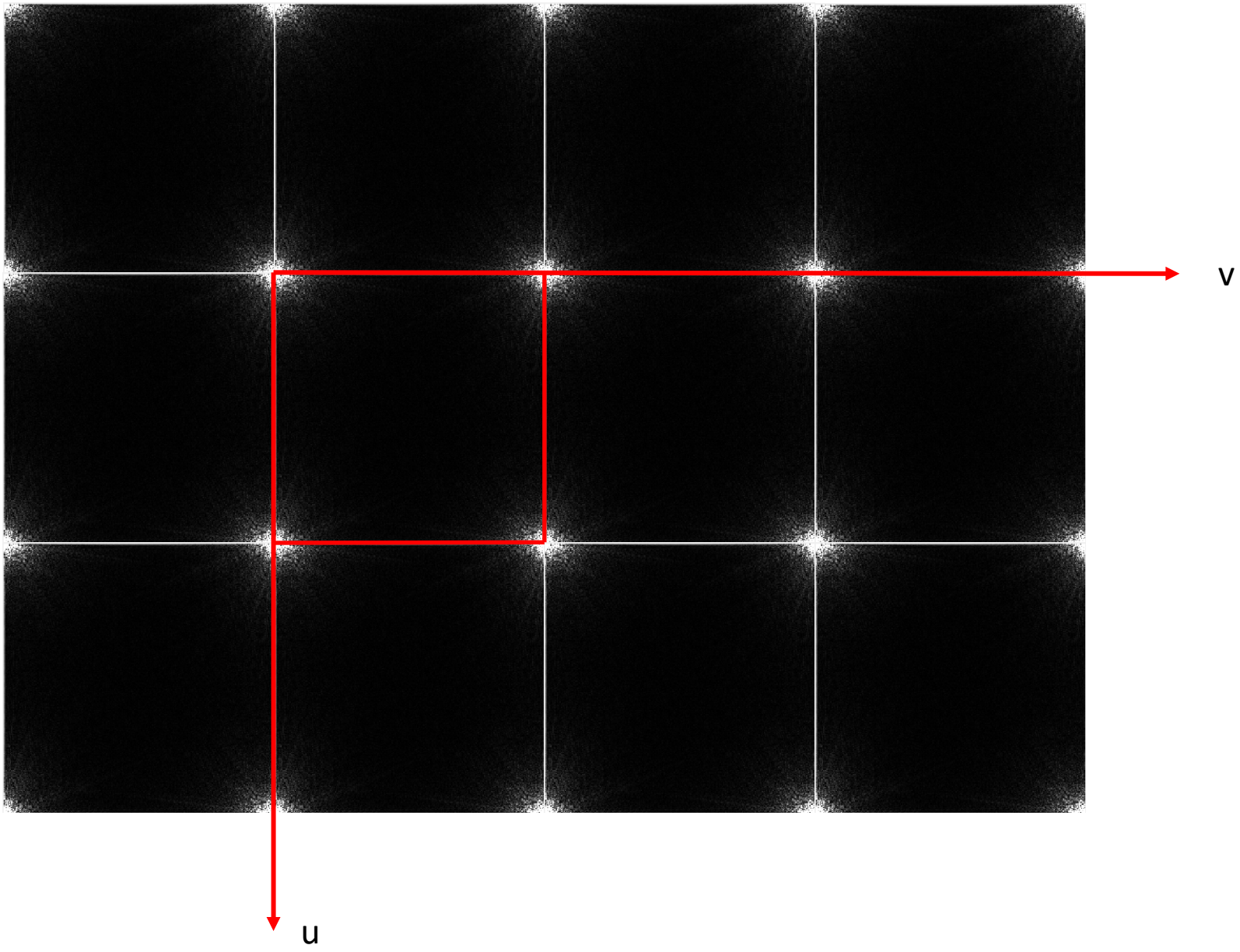


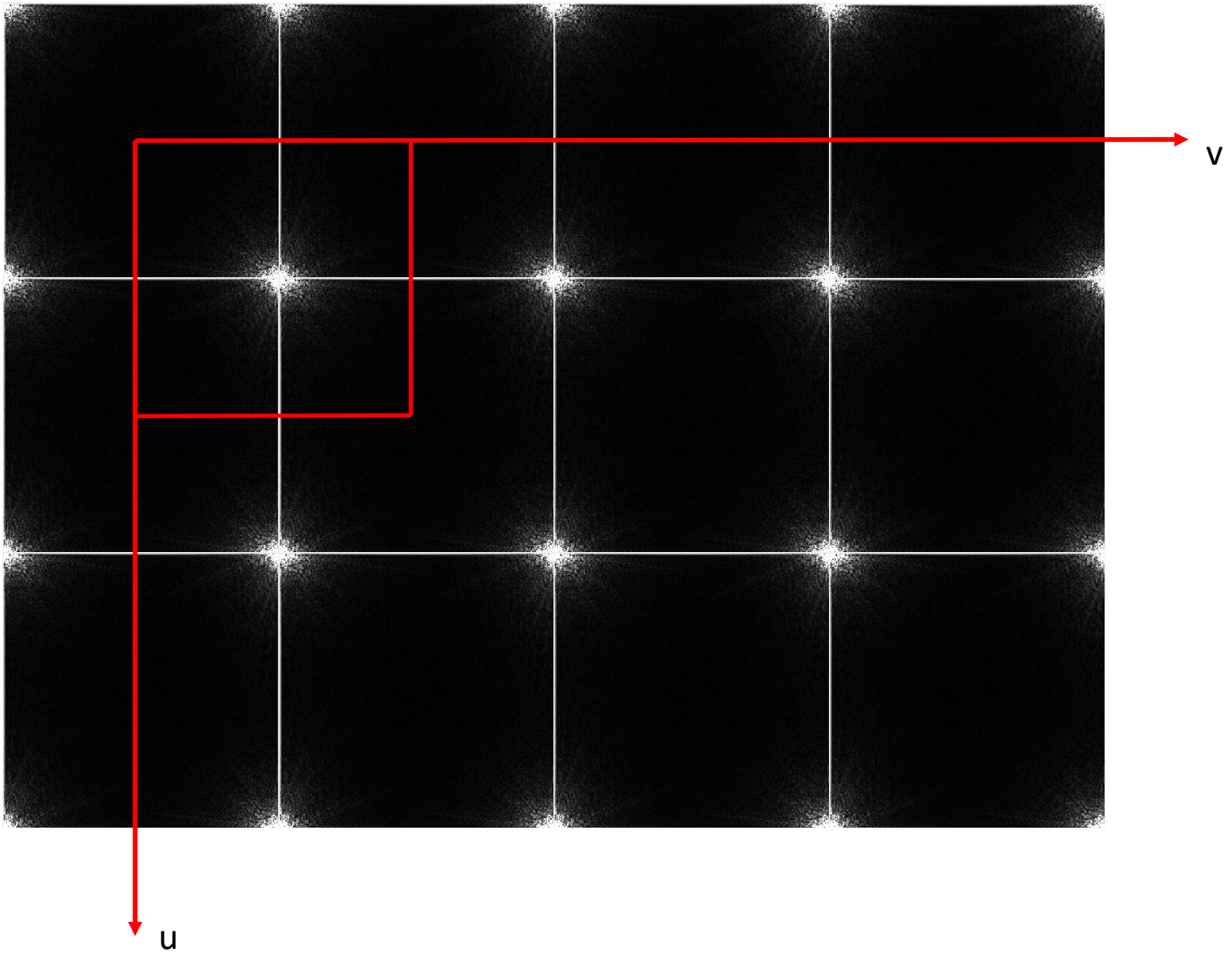
频谱图



频谱搬移

频谱图中间部分为低频部分，越靠外边频率越高。图像中的能量主要集中在低频区，高频能量很少或为零。


v
u


v
u

4.6.2 Translation and rotation

$$f(x,y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u-u_0,v-v_0)$$

- Translation

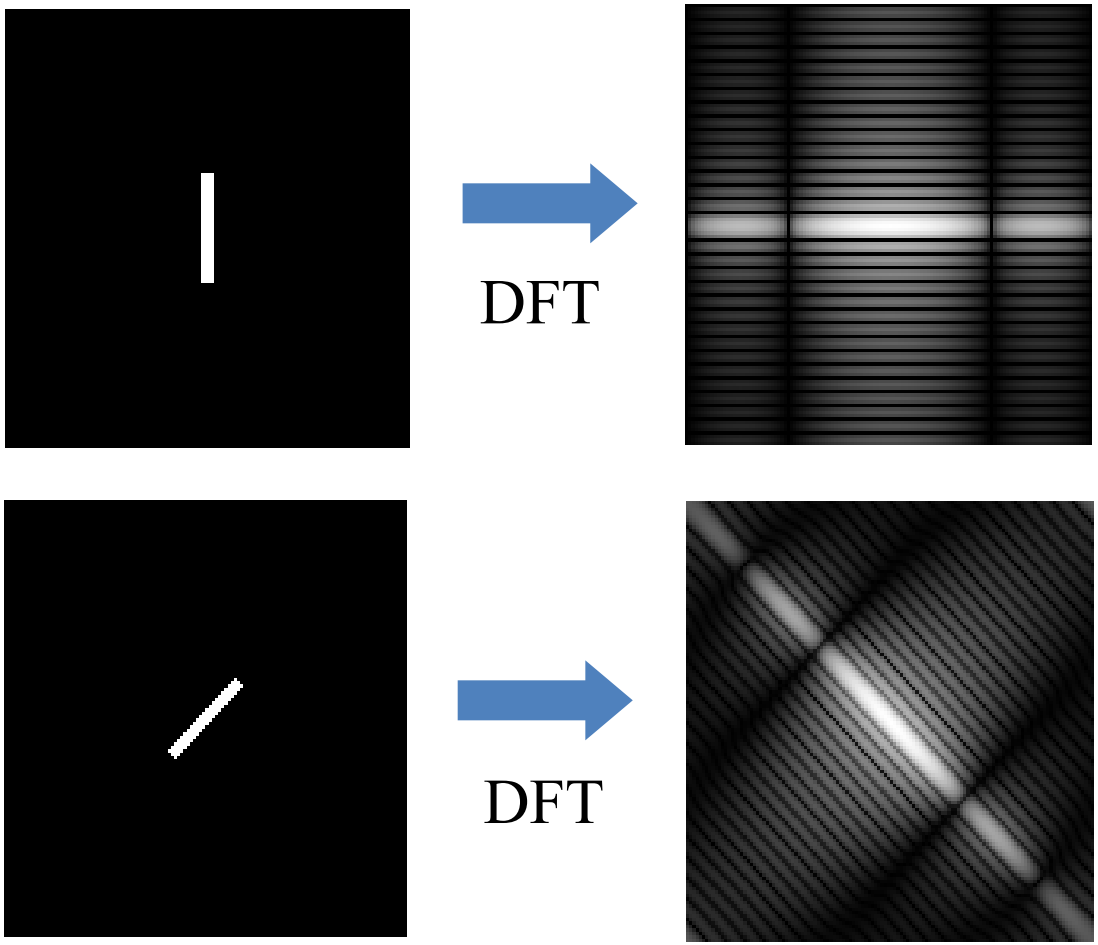
$$f(x-x_0,y-y_0) \Leftrightarrow F(u,v)e^{-j2\pi(x_0u/M+y_0v/N)}$$

- Using the polar coordinates:

$$f(r,\theta+\theta_0) \Leftrightarrow F(\omega,\varphi+\theta_0)$$

Rotating f(x,y) rotates F(u,v) by the same angle.

The Property of Two-Dimensional DFT Rotation



$$f(x,y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u-u_0,v-v_0)$$

$$f(x,y)(-1)^{x+y} \Leftrightarrow F(u-\frac{M}{2},v-\frac{N}{2})$$

- This equation shifts the data so that F(0,0) is at the center of the frequency rectangle defined by the intervals [0,M-1] and [0,N-1].

Fourier spectrum and phase angle

- In general, the 2D DFT is complex, it can be expressed in polar form:

$$F(u,v) = |F(u,v)| e^{j\phi(u,v)}$$

The magnitude is called the Fourier spectrum

$$|F(u,v)| = [R^2(u,v) + I^2(u,v)]^{\frac{1}{2}}$$

Fourier spectrum and phase angle

- The phase angle

$$\phi(u,v) = \arctan\left[\frac{I(u,v)}{R(u,v)}\right]$$

The power spectrum

$$P(u,v) = |F(u,v)|^2 = R^2(u,v) + I^2(u,v)$$

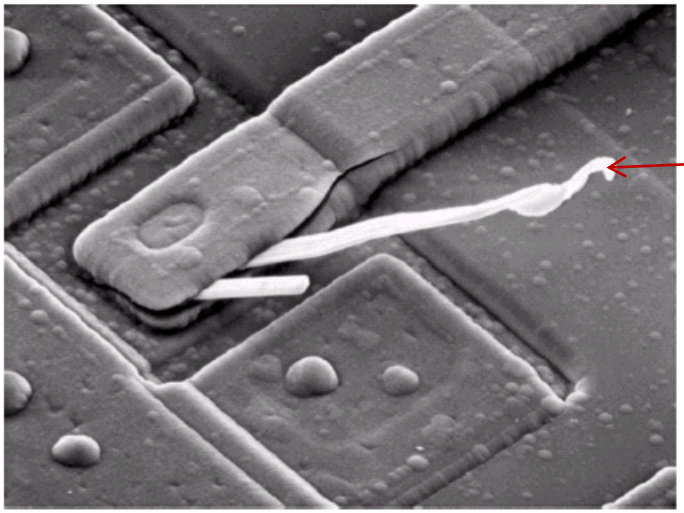
Fourier spectrum and phase angle

$$F(0,0) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)$$

$$= MN\frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y) = MN\,\overline{f}(x,y)$$

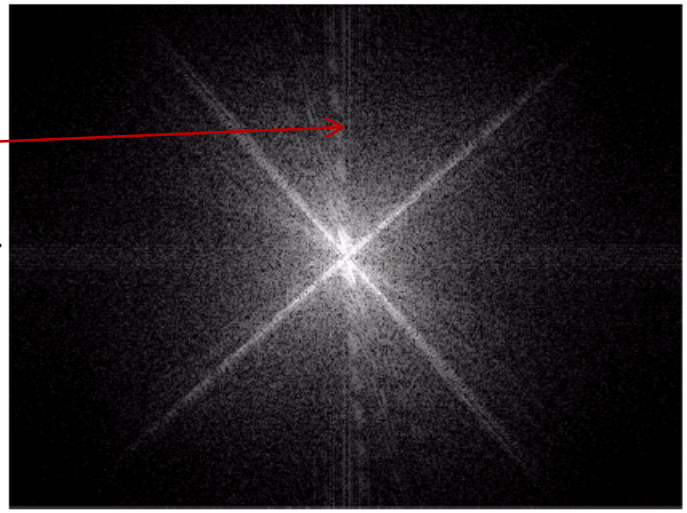
The zero-frequency term is proportional to the average value of $f(x,y)$. Because the constant MN Usually is large, |F(0,0)| typically is the largest component of the spectrum. Called DC component.

DFT & Images (cont…)



Scanning electron microscope image of an integrated circuit magnified ~2500 times

DFT



Fourier spectrum of the image

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

separability可分性

$$F(u,v)=\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}f(x,y)\exp[-\mathrm{j}2\pi(ux/M+vy/N)]$$

$$u=0,1,\cdots,\mathrm{M}-1$$
$$v=0,1,\cdots,N-1$$

row

$$F(x,v)=\left[\sum_{y=0}^{N-1}f(x,y)\exp[-\mathrm{j}2\pi vy/N]\right] \qquad v=0,1,\cdots,N-1$$

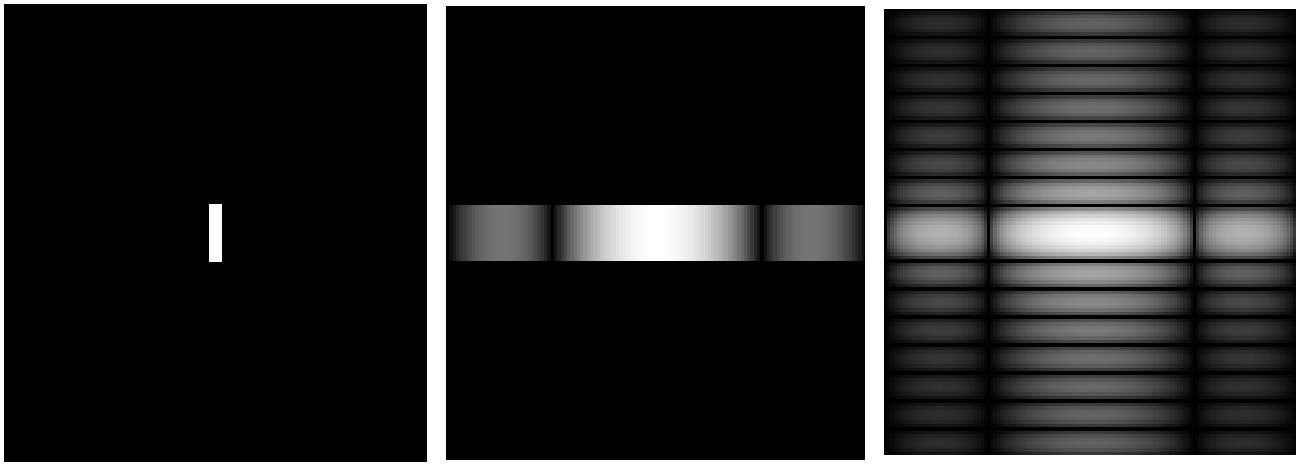
column

$$F(u,v)=\sum_{x=0}^{M-1}F(x,v)\exp[-\mathrm{j}2\pi u\mathrm{x}/M]$$

$$u=0,1,\cdots,\mathrm{M}-1$$
$$v=0,1,\cdots,N-1$$

The 2D DFT **F(u,v)** can be obtained by

1. taking the 1D DFT of every row of image **f(x,y)**, **F(u,y)**,
2. taking the 1D DFT of every column of **F(u,v)**



(a)f(x,y) (b)F(u,y) (c)F(u,v)

Property	Expression(s)
Fourier transform	$F(u,v) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(ux/M+vy/N)}$
Inverse Fourier transform	$f(x,y) = \sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)e^{j2\pi(ux/M+vy/N)}$
Polar representation	$F(u,v) = \lvert F(u,v)\rvert e^{-j\phi(u,v)}$
Spectrum	$\lvert F(u,v)\rvert = \left[R^2(u,v) + I^2(u,v)\right]^{1/2}$, $R = \text{Real}(F)$ and $I = \text{Imag}(F)$
Phase angle	$\phi(u,v) = \tan^{-1}\left[\frac{I(u,v)}{R(u,v)}\right]$
Power spectrum	$P(u,v) = \lvert F(u,v)\rvert^2$
Average value	$\bar{f}(x,y) = F(0,0) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)$
Translation	$f(x,y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u-u_0, v-v_0)$ $f(x-x_0, y-y_0) \Leftrightarrow F(u,v)e^{-j2\pi(ux_0/M+vy_0/N)}$ When $x_0 = u_0 = M/2$ and $y_0 = v_0 = N/2$, then $f(x,y)(-1)^{x+y} \Leftrightarrow F(u-M/2, v-N/2)$ $f(x-M/2, y-N/2) \Leftrightarrow F(u,v)(-1)^{u+v}$

4.7 The basic of filtering in the frequency domain

- Convolution theorem:
  - The discrete convolution of two functions *f*(*x*,*y*) and *h*(*x*,*y*) of size *M*X*N* is defined as

$$f(x,y)*h(x,y)=\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}f(m,n)h(x-m,y-n)$$

  - Let *F*(*u*,*v*) and *H*(*u*,*v*) denote the Fourier transforms of *f*(*x*,*y*) and *h*(*x*,*y*), then

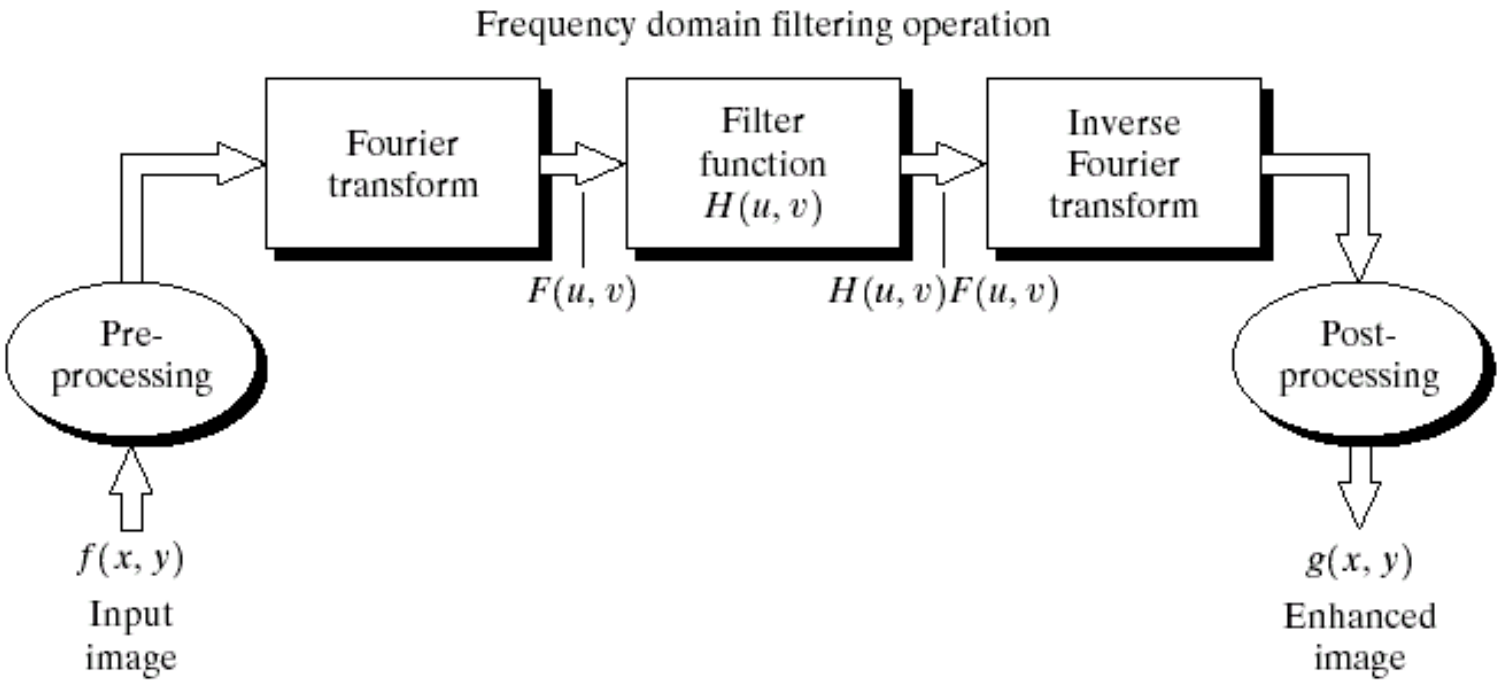
$$f(x,y)*h(x,y)\Leftrightarrow F(u,v)H(u,v)$$

$$f(x,y)h(x,y)\Leftrightarrow\frac{1}{MN}F(u,v)*H(u,v)$$

The basic of filtering in the frequency domain

To filter an image in the frequency domain:

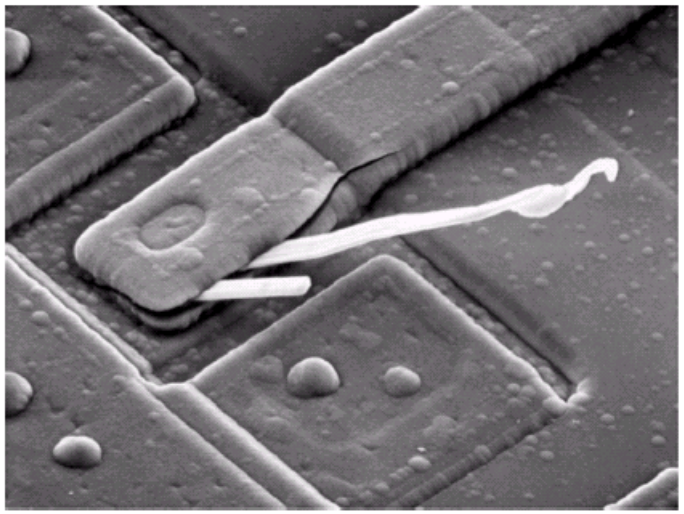
1. Compute $F(u,v)$ the DFT of the image
2. Multiply $F(u,v)$ by a filter function $H(u,v)$
3. Compute the inverse DFT of the result

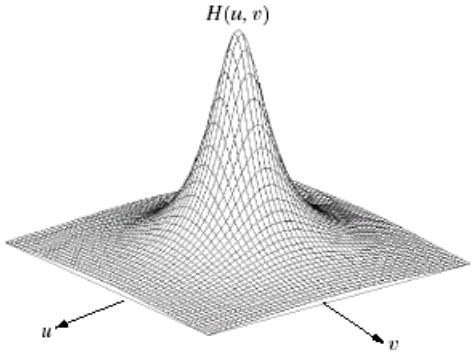


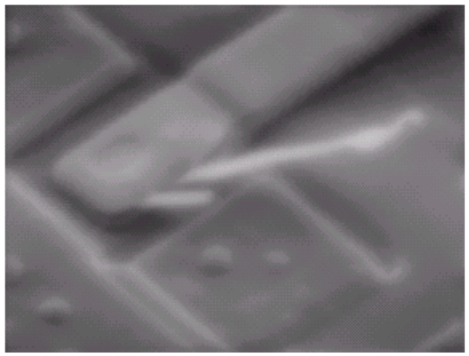
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

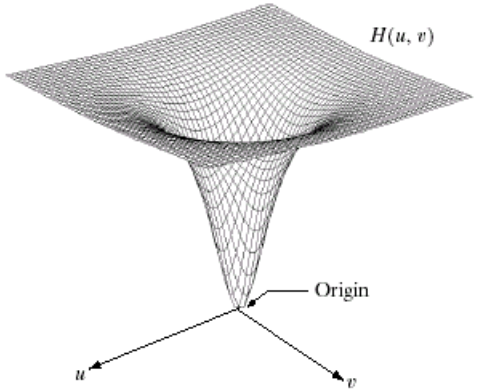
Some Basic Frequency Domain Filters

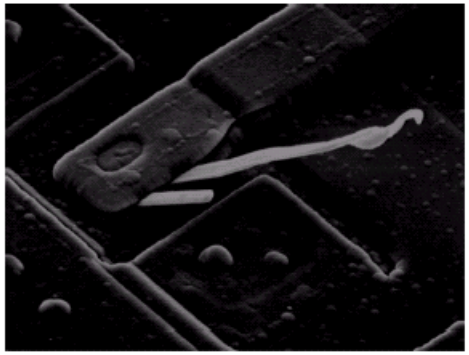
Low Pass Filter







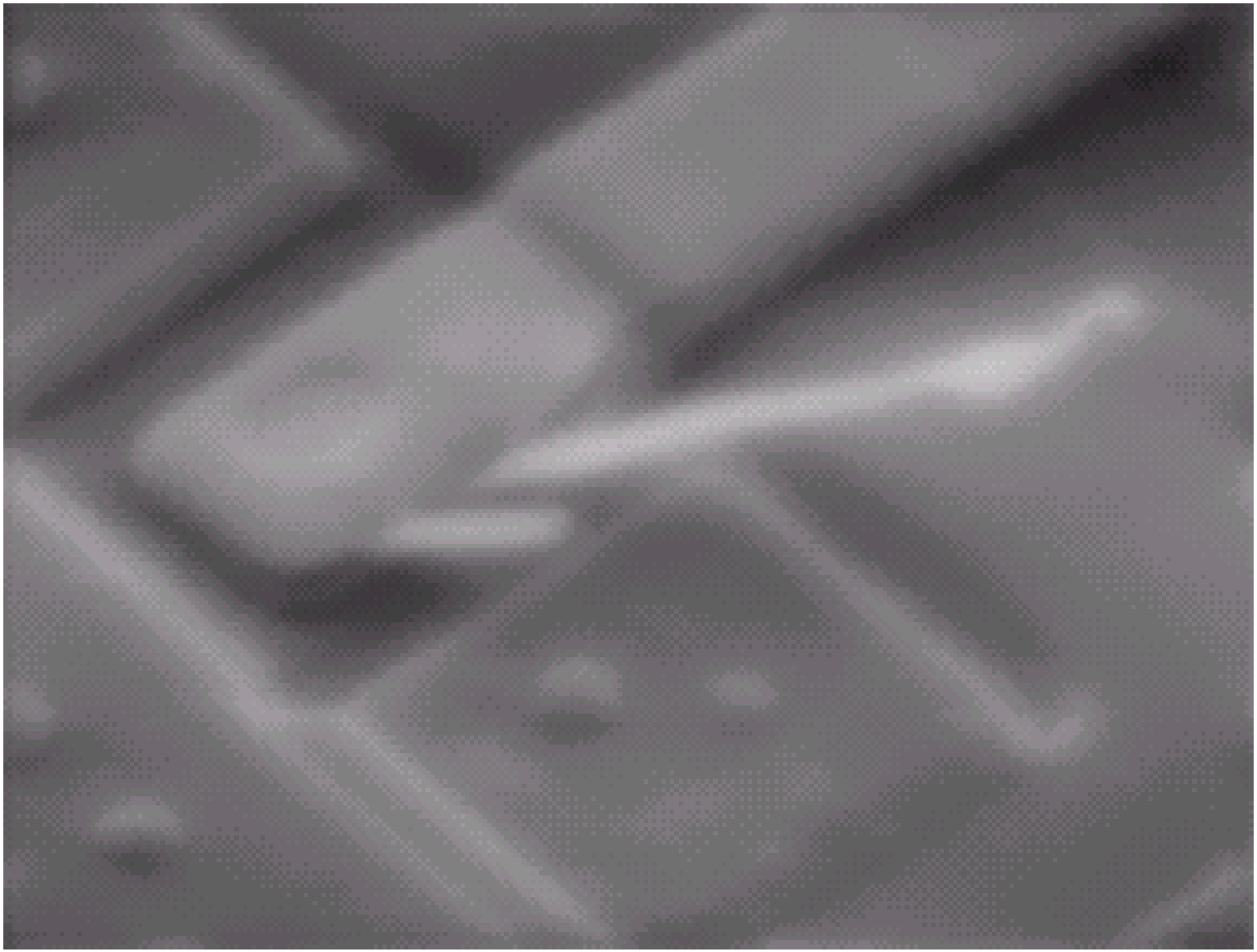




High Pass Filter

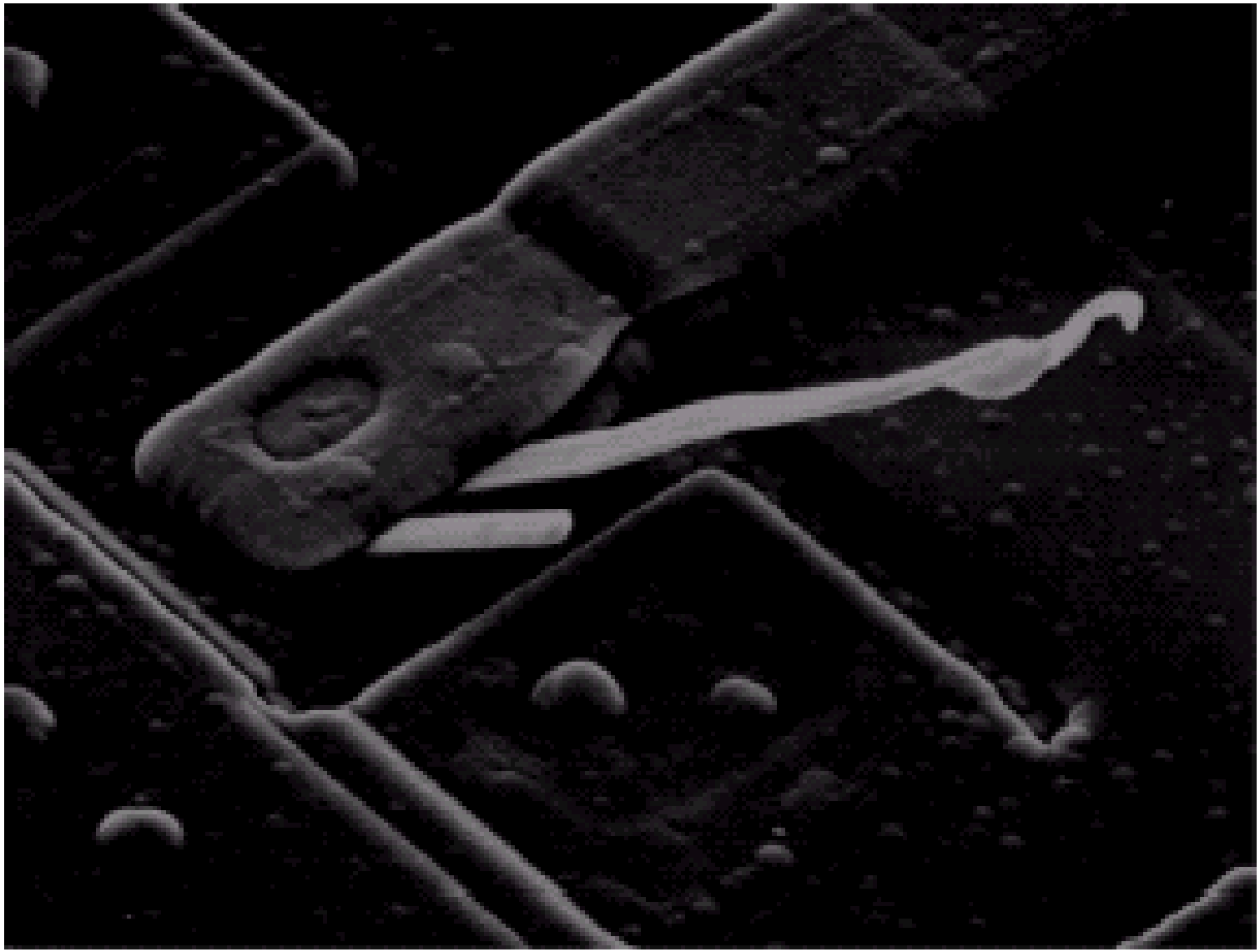
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Some Basic Frequency Domain Filters



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Some Basic Frequency Domain Filters



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Ideal Low Pass Filter (cont…)

The transfer function for the ideal low pass filter can be given as:

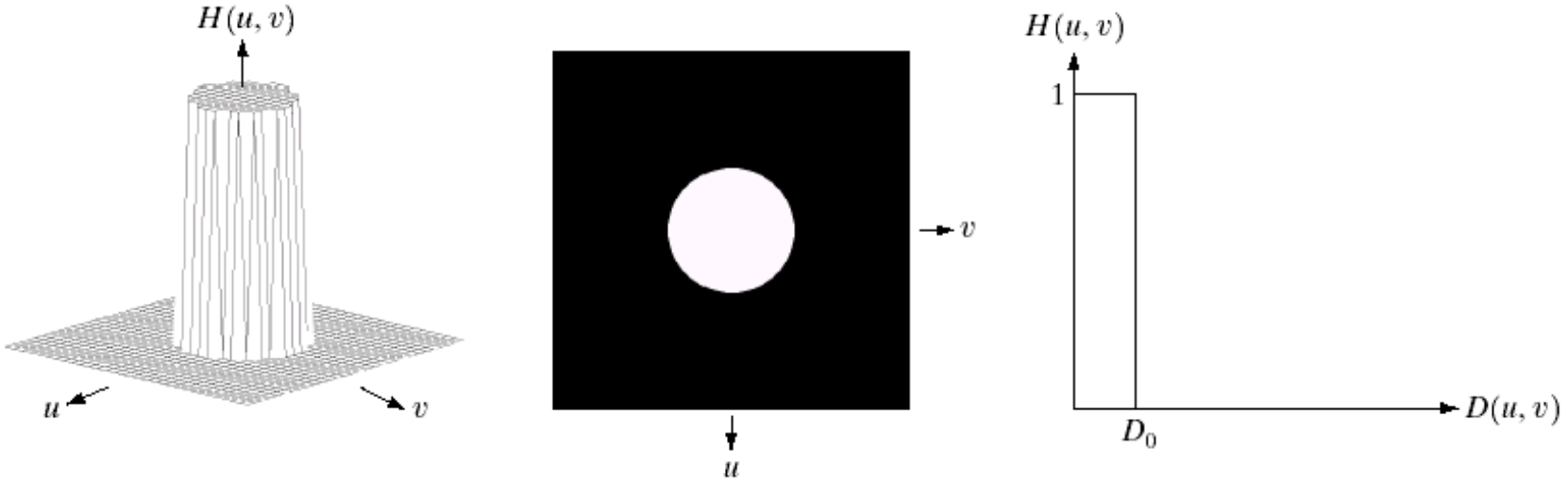
$$H(u,v) = \begin{cases} 1 \text{ if } D(u,v) \le D_0 \\ 0 \text{ if } D(u,v) > D_0 \end{cases}$$

where $D(u,v)$ is given as:

$$D(u,v) = [(u - M/2)^2 + (v - N/2)^2]^{1/2}$$

Ideal Low Pass Filter

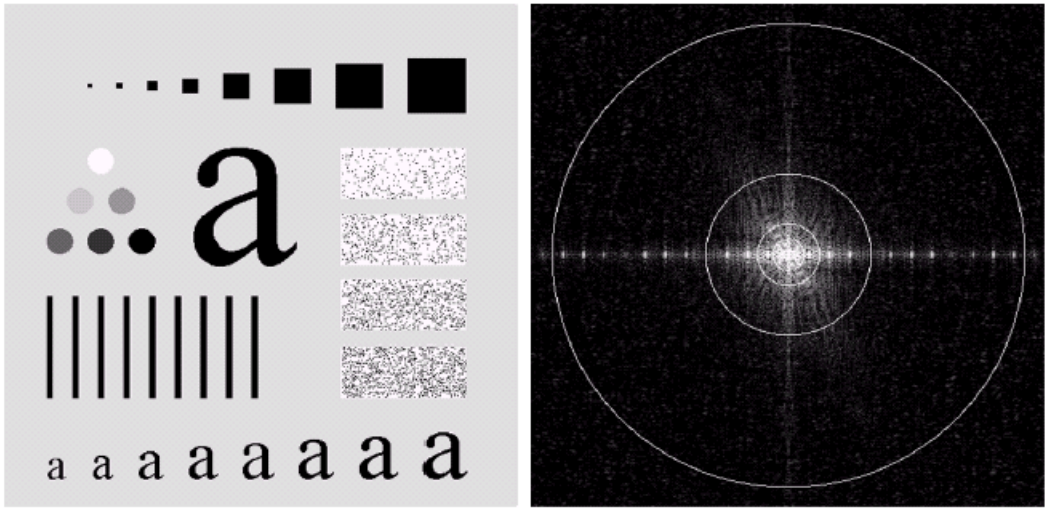
Simply cut off all high frequency components that are a specified distance D_0 from the origin of the transform



changing the distance changes the behaviour of the filter

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Ideal Low Pass Filter (cont…)



Above we show an image, it's Fourier spectrum and a series of ideal low pass filters of radius 5, 15, 30, 80 and 230 superimposed on top of it

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Expanded knowledge

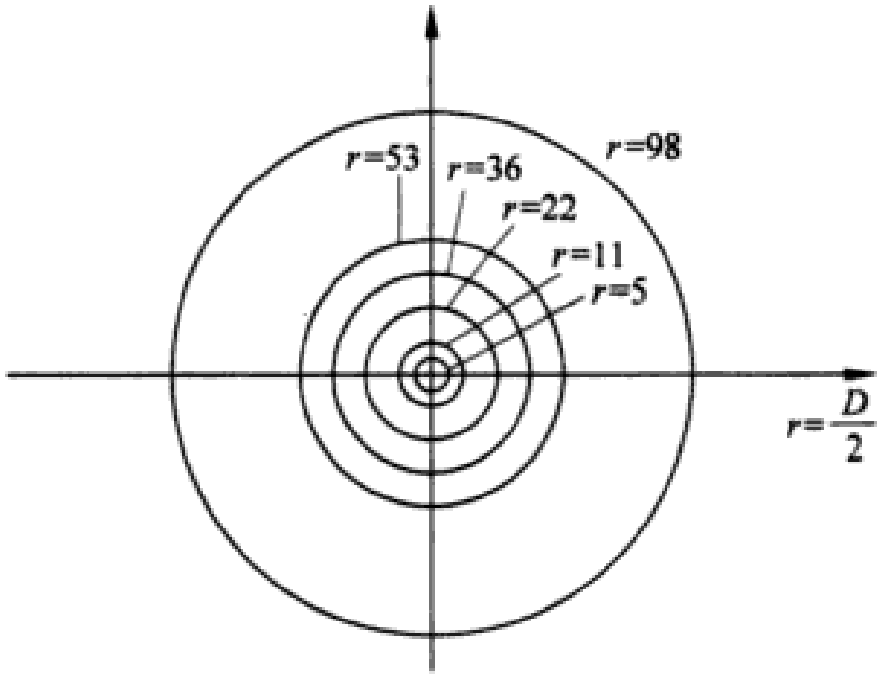


图 3-24　低通滤波的能量和 D_0 的关系

表 3-6　半径 r 与包含总能量的关系

半径 r	包含总能量%	半径 r	包含总能量%
5	90.0	36	99.0
11	96.0	53	99.5
22	98.0	98	99.9

理想低通滤波器

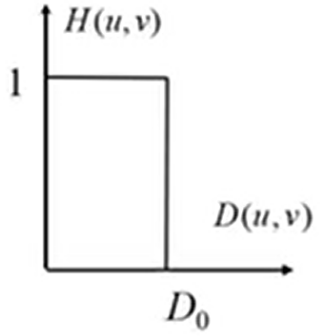
特点：

- 物理上不可实现
- 滤除高频成分使图象变模糊

振铃现象明显

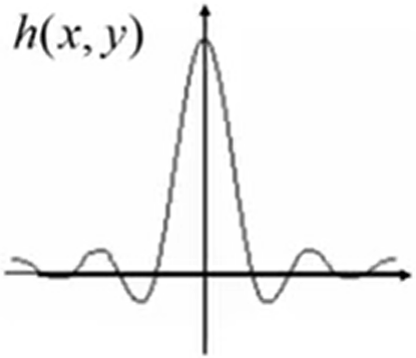
理想低通滤波的振铃现象

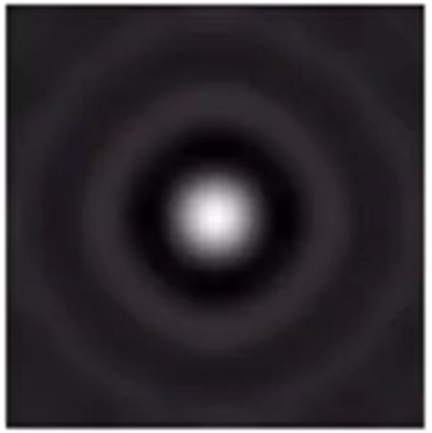
$$H(u,v)=\begin{cases}1 & D(u,v)\le D_0\\ 0 & D(u,v)>D_0\end{cases}$$





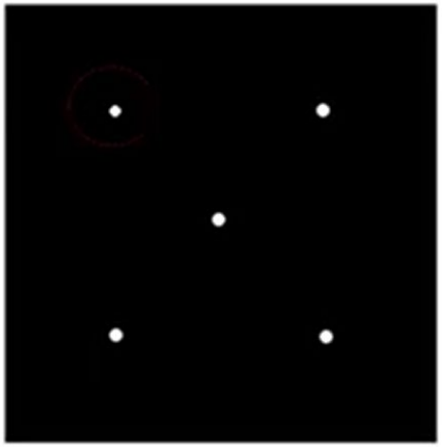
频域投影





空域投影

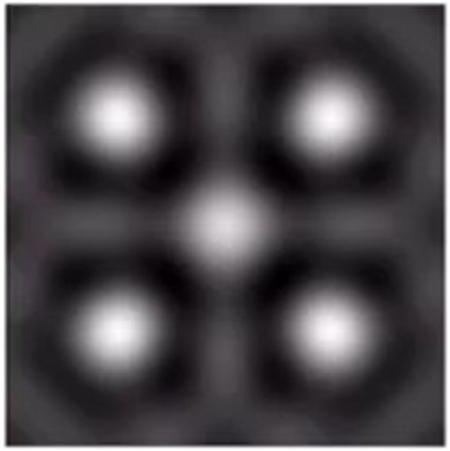
理想低通滤波的振铃现象



原图（五个小点）



理想低通滤波

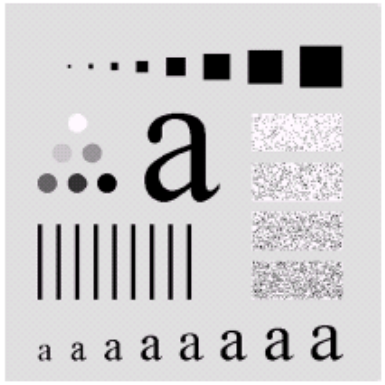


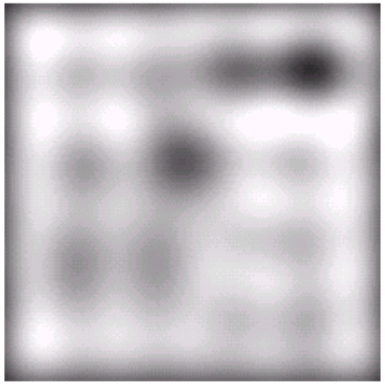
滤波后图

（振铃现象）

Ideal Low Pass Filter (cont…)

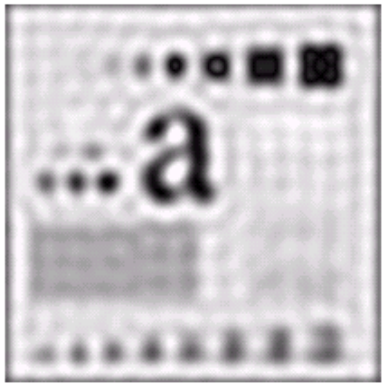
Original image

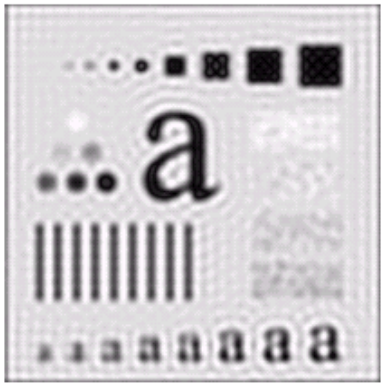




Result of filtering with ideal low pass filter of radius 5

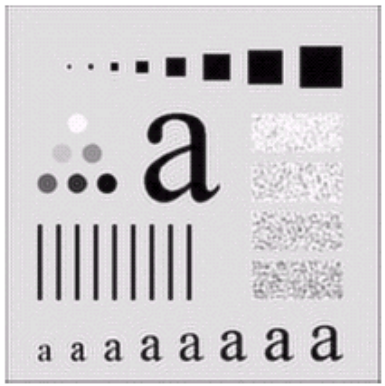
Result of filtering with ideal low pass filter of radius 15

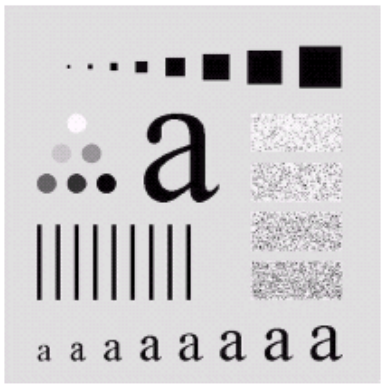




Result of filtering with ideal low pass filter of radius 30

Result of filtering with ideal low pass filter of radius 80

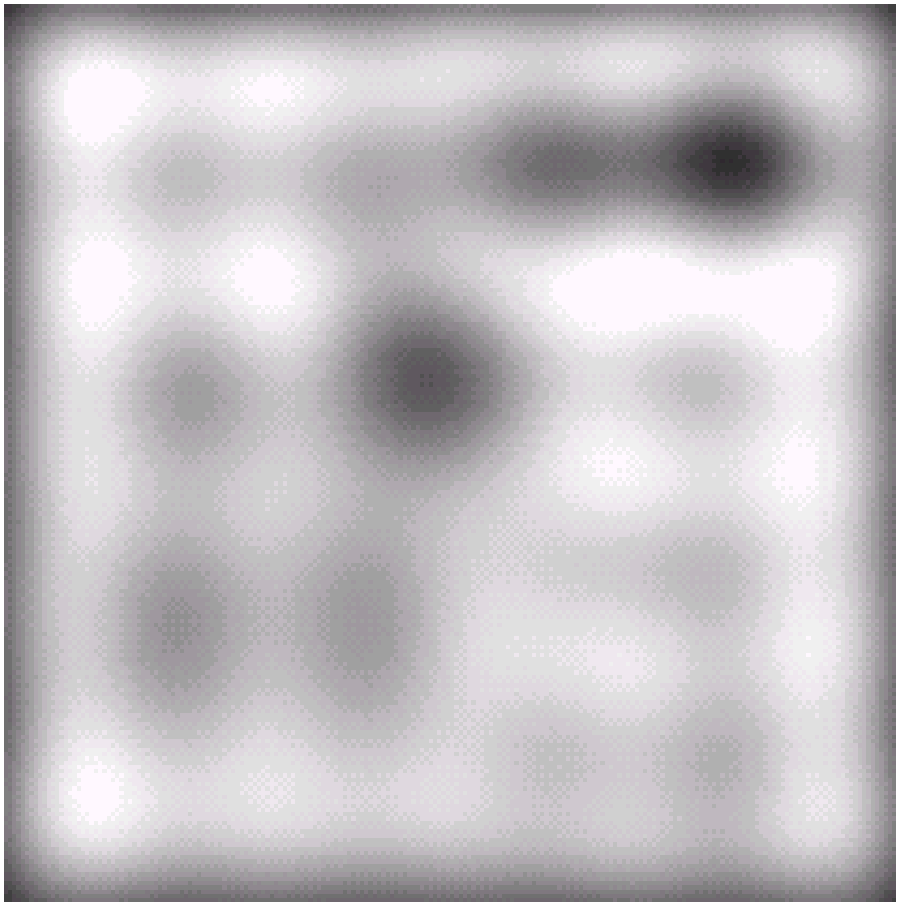




Result of filtering with ideal low pass filter of radius 230

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

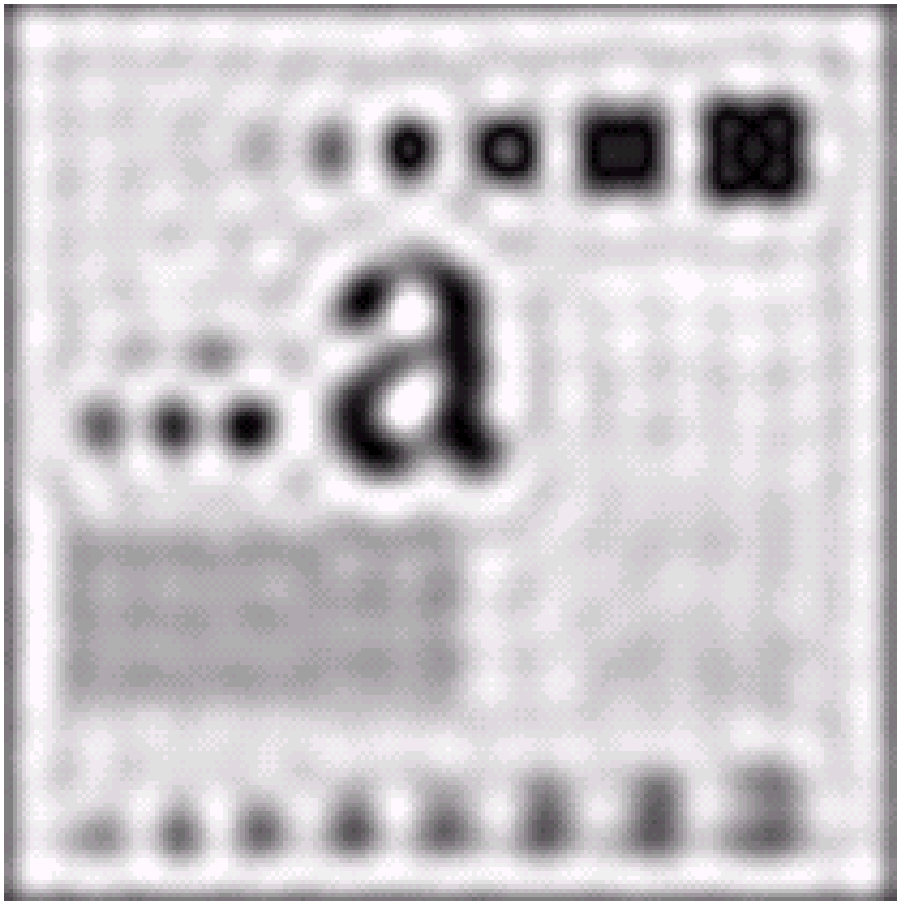
Ideal Low Pass Filter (cont…)



Result of filtering with ideal low pass filter of radius 5

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Ideal Low Pass Filter (cont…)



Result of filtering with ideal low pass filter of radius 15

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

单选题 25分

设置

The key concept of Fourier transformation is any function that periodically repeats itself can be expressed as a sum of sines and cosines of different frequencies each multiplied by a different coefficient.

A Right

B Wrong

提交

单选题 25分

设置

The origin frequency component of the image Fourier transformation locate at the ( ) of frequency domain.

A Center

B Four corners

C Random locations

提交

单选题 25分

设置

The high frequency components relate ( )of objects in the image

A Edges or abrupt intensity changes

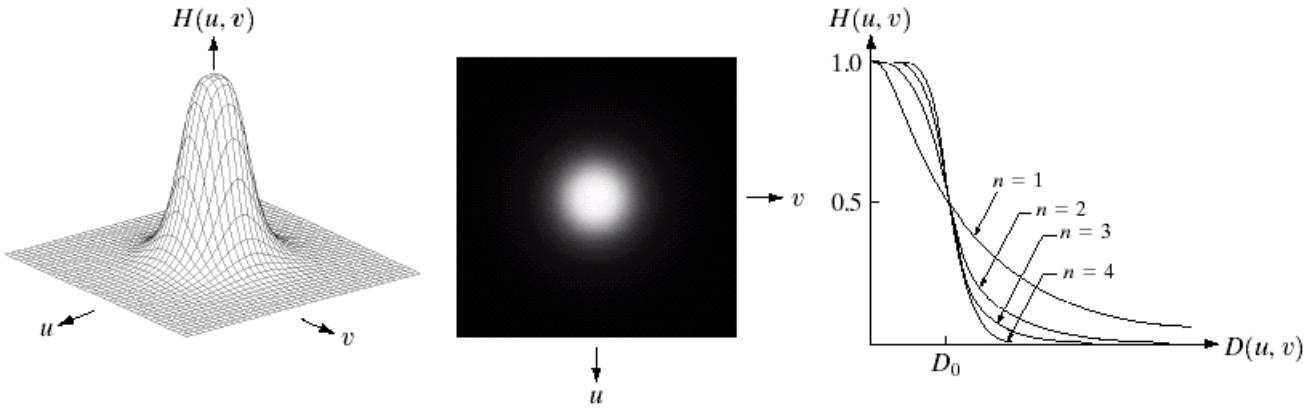
B Smooth intensity variations

提交

Butterworth Lowpass Filters

The transfer function of a Butterworth lowpass filter of order n with cutoff frequency at distance D_0 from the origin is defined as:

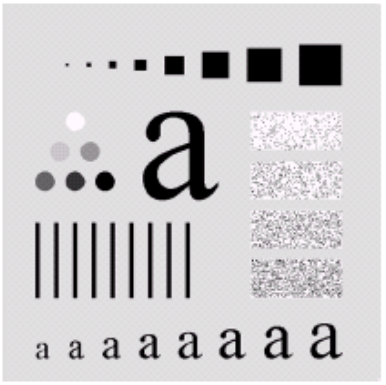
$$H(u,v) = \frac{1}{1+[D(u,v)/D_0]^{2n}}$$

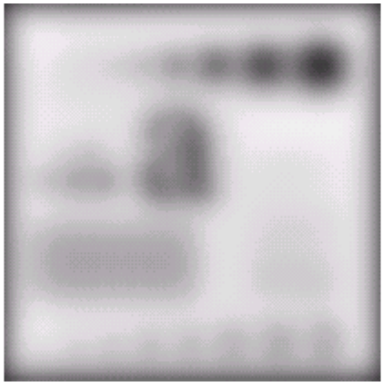


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Butterworth Lowpass Filter (cont…)

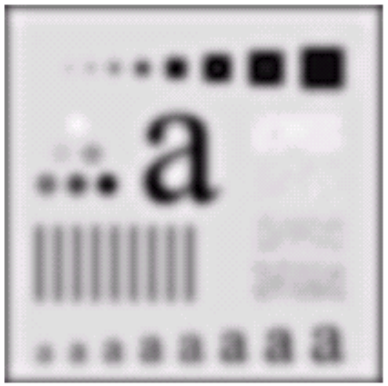
Original image

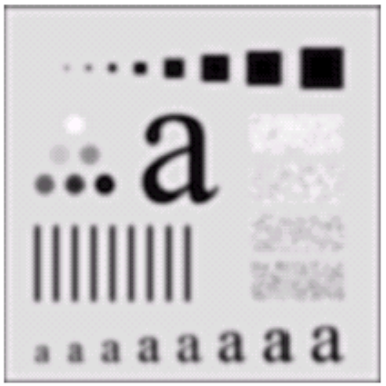




Result of filtering with Butterworth filter of order 2 and cutoff radius 5

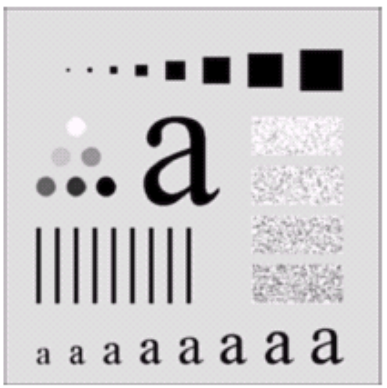
Result of filtering with Butterworth filter of order 2 and cutoff radius 15

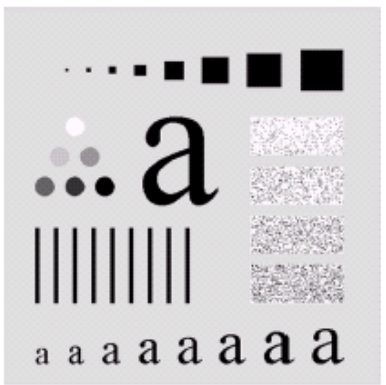




Result of filtering with Butterworth filter of order 2 and cutoff radius 30

Result of filtering with Butterworth filter of order 2 and cutoff radius 80

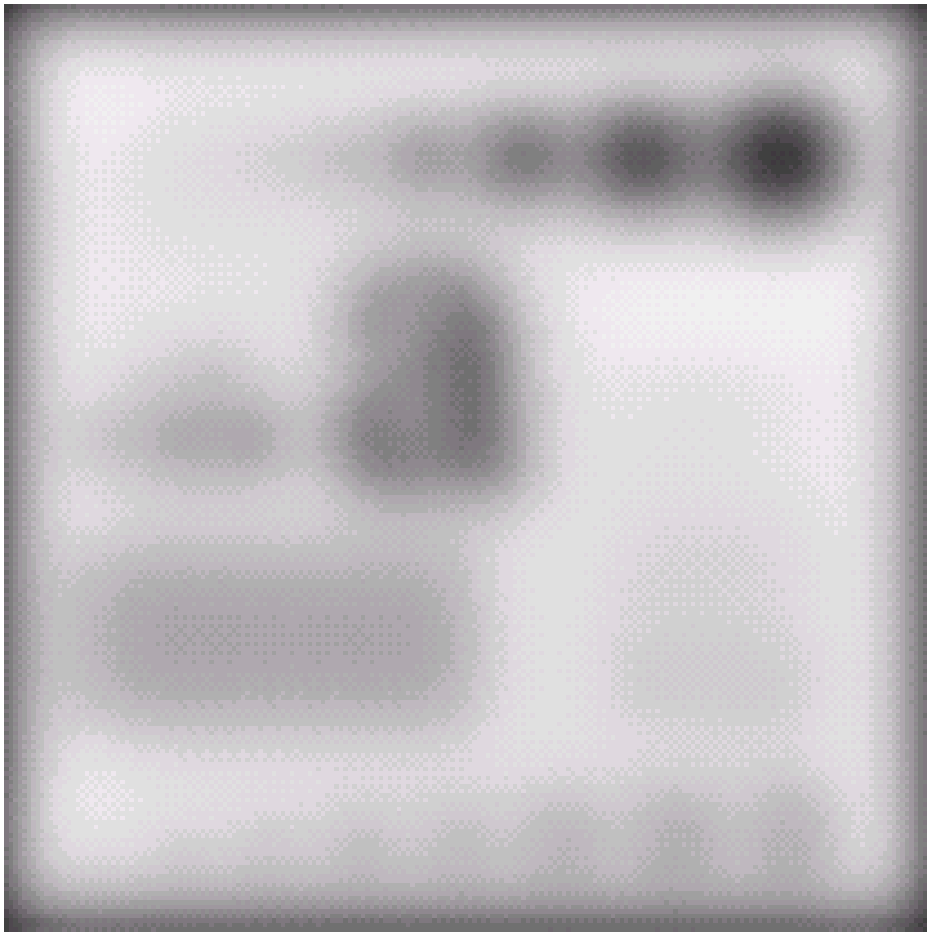




Result of filtering with Butterworth filter of order 2 and cutoff radius 230

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Butterworth Lowpass Filter (cont…)

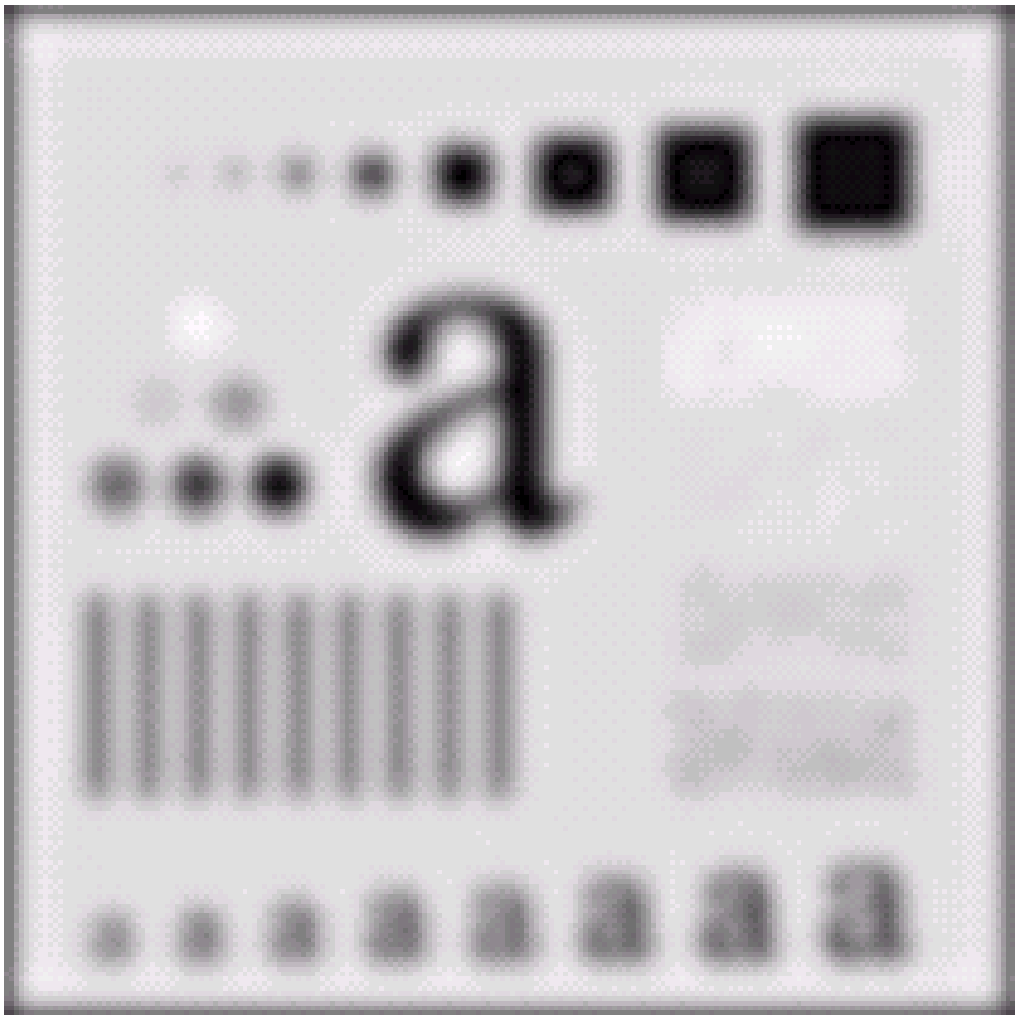


Result of filtering with Butterworth filter of order 2 and cutoff radius 5

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

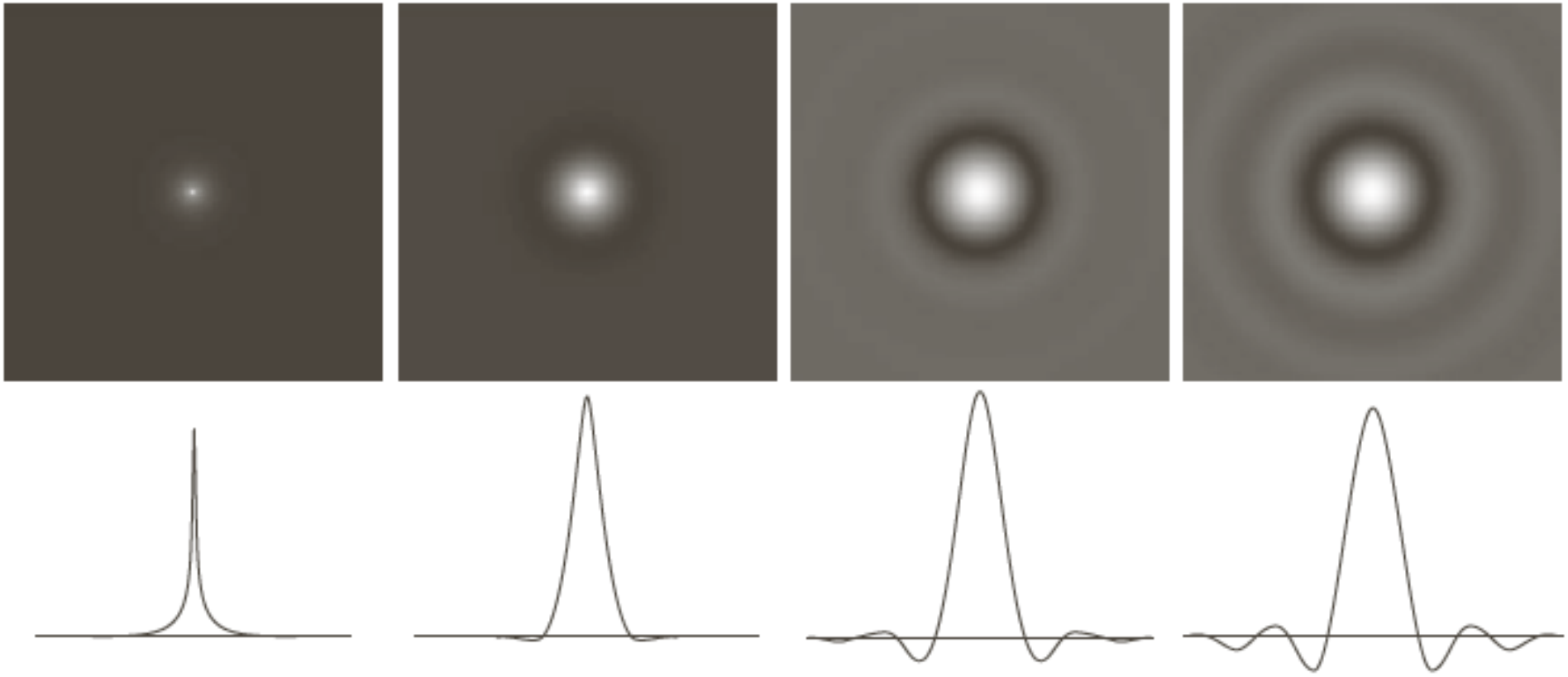
Butterworth Lowpass Filter (cont…)

Result of filtering with Butterworth filter of order 2 and cutoff radius 15



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Difference order BLPFs



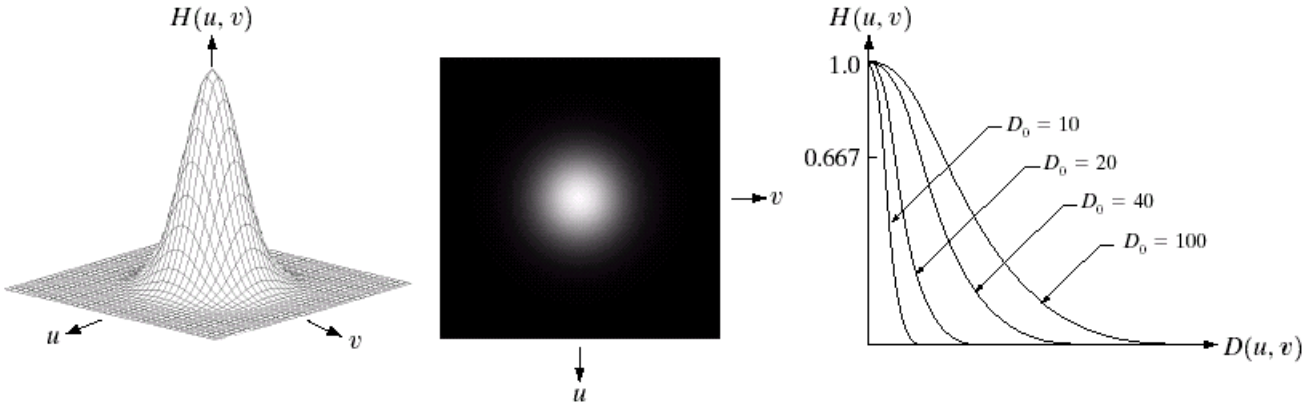
a b c d

FIGURE 4.46 (a)–(d) Spatial representation of BLPFs of order 1, 2, 5, and 20, and corresponding intensity profiles through the center of the filters (the size in all cases is 1000 × 1000 and the cutoff frequency is 5). Observe how ringing increases as a function of filter order.

4.8.3 Gaussian Lowpass Filters

The transfer function of a Gaussian lowpass filter is defined as:

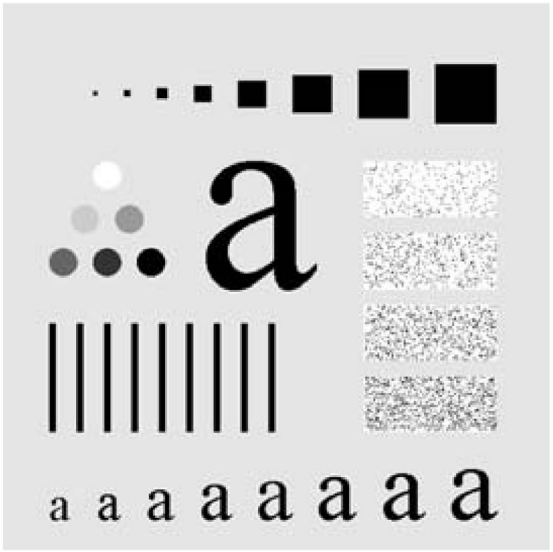
$$H(u,v) = e^{-D^2(u,v)/2D_0{}^2}$$

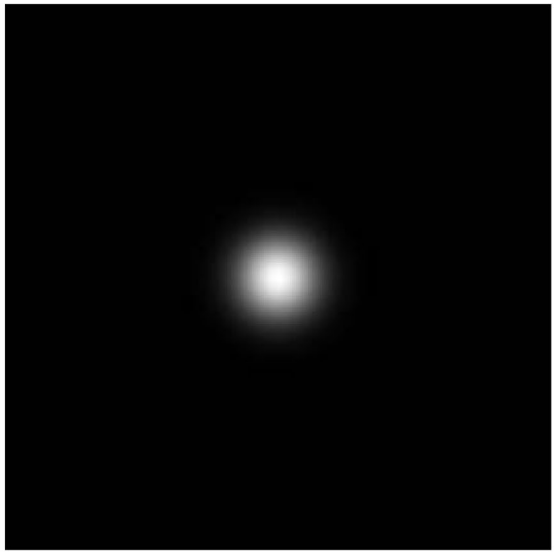


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Lowpass Filtering Examples (cont…)

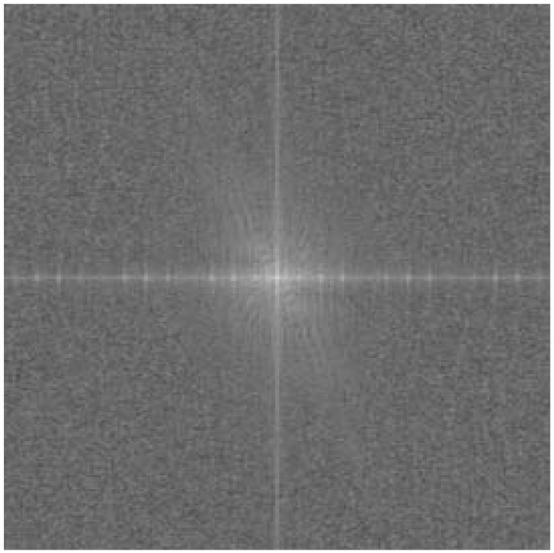
Original image





Gaussian lowpass filter

Spectrum of original image



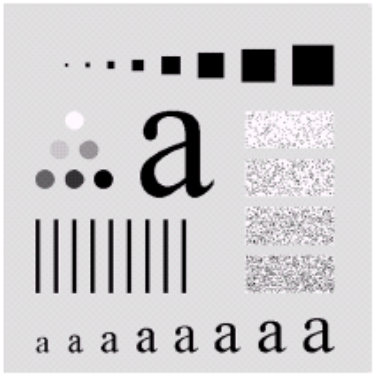


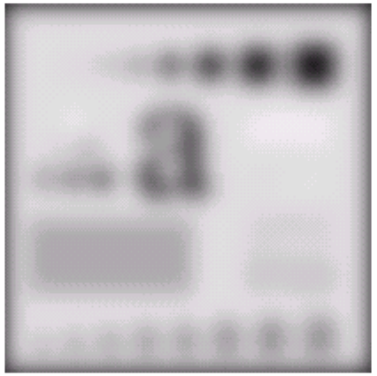
Processed image

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Gaussian Lowpass Filters (cont…)

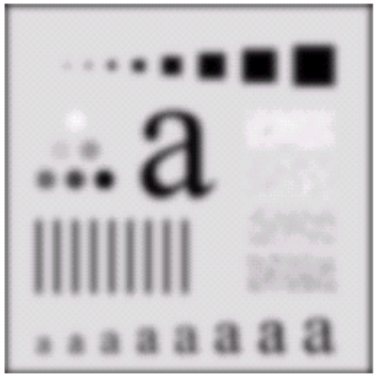
Original image

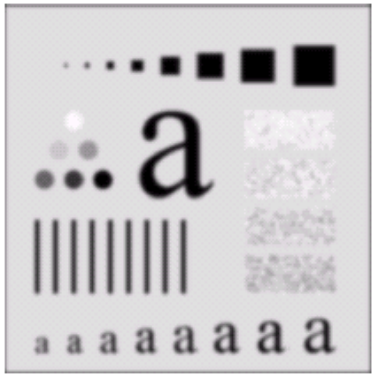




Result of filtering with Gaussian filter with cutoff radius 5

Result of filtering with Gaussian filter with cutoff radius 15

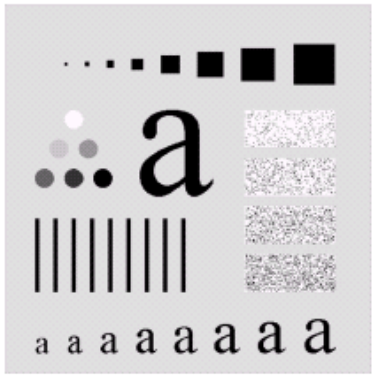




Result of filtering with Gaussian filter with cutoff radius 30

Result of filtering with Gaussian filter with cutoff radius 85



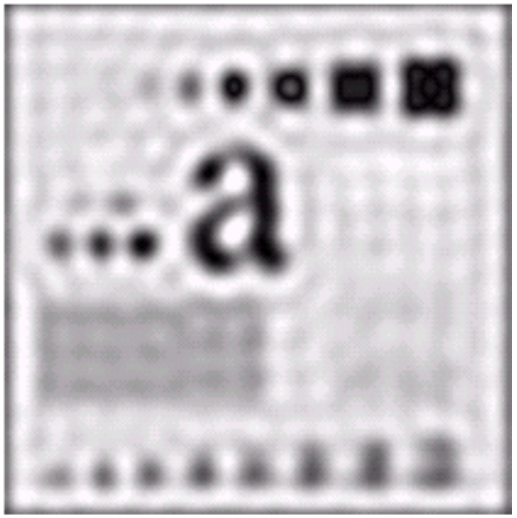


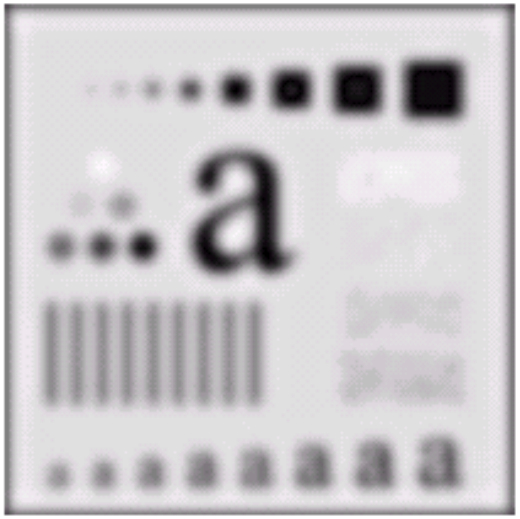
Result of filtering with Gaussian filter with cutoff radius 230

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Lowpass Filters Compared

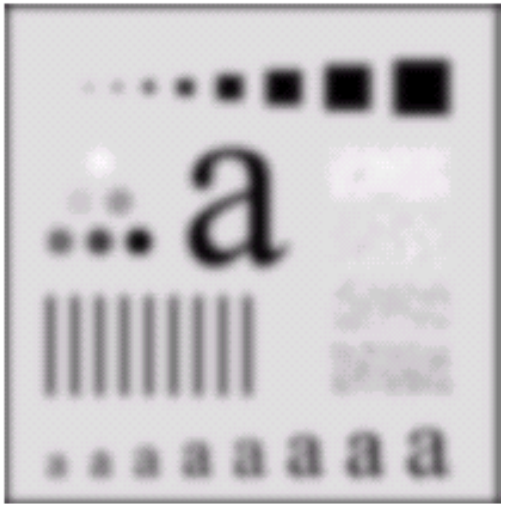
Result of filtering with ideal low pass filter of radius 15





Result of filtering with Butterworth filter of order 2 and cutoff radius 15

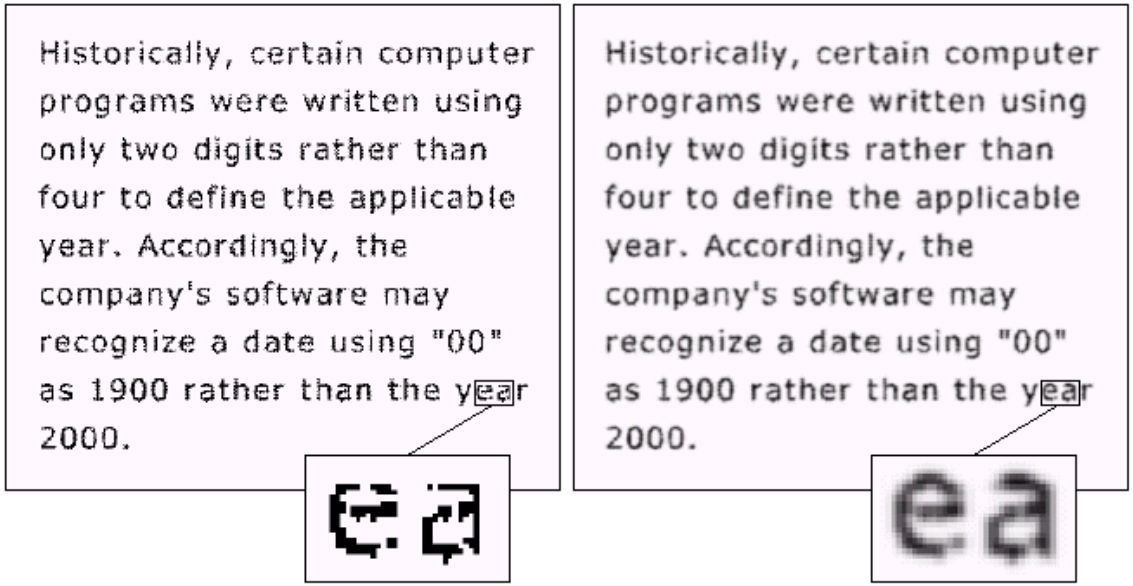
Result of filtering with Gaussian filter with cutoff radius 15



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Lowpass Filtering Examples

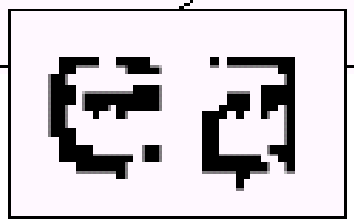
A low pass Gaussian filter is used to connect broken text



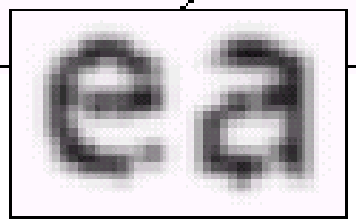
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Lowpass Filtering Examples

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Lowpass Filtering Examples (cont…)

Different lowpass Gaussian filters used to remove blemishes in a photograph



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Lowpass Filtering Examples (cont…)



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

4.9 Sharpening in the Frequency Domain

Edges and fine detail in images are associated with high frequency components

High pass filters – only pass the high frequencies, drop the low ones

High pass frequencies are precisely the reverse of low pass filters, so:

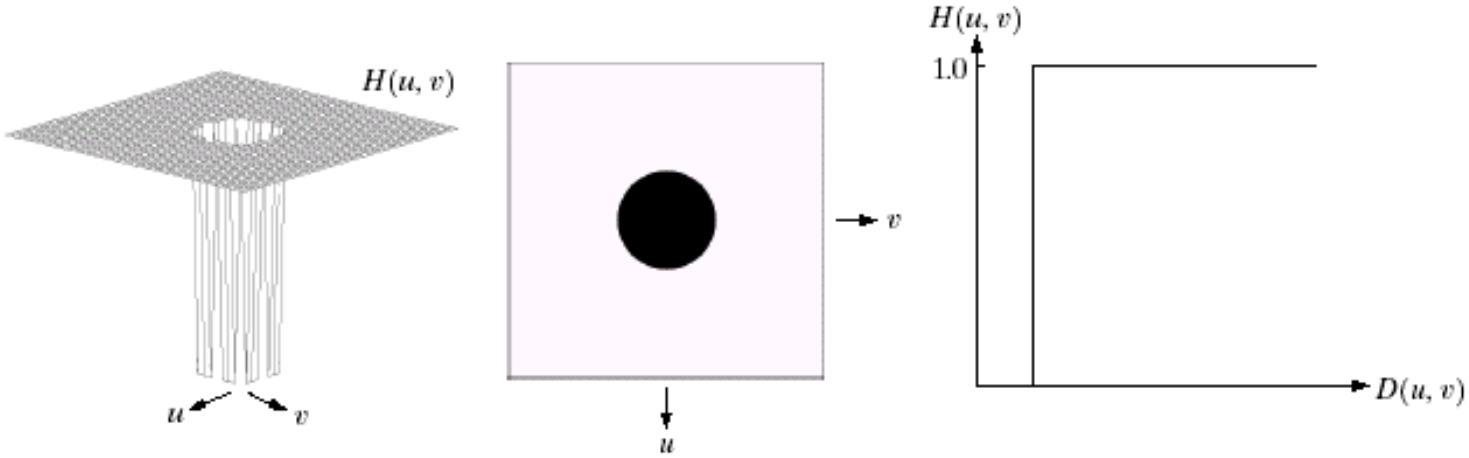
$$H_{hp}(u, v) = 1 - H_{lp}(u, v)$$

4.9.1 Ideal High Pass Filters

The ideal high pass filter is given as:

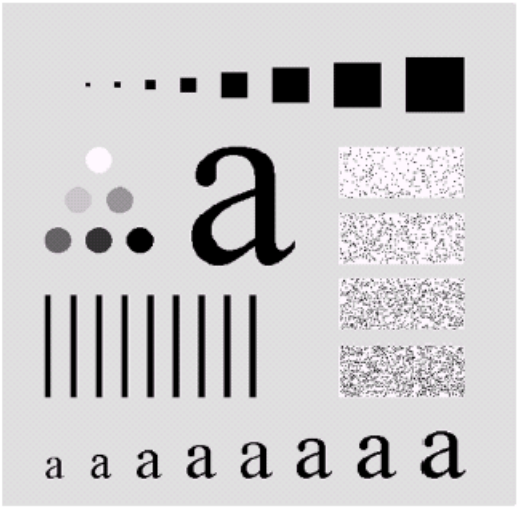
$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \le D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$$

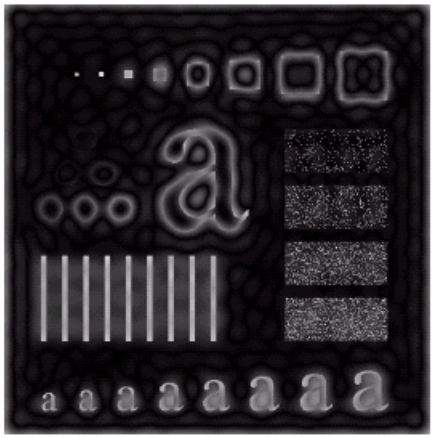
where D_0 is the cut off distance as before



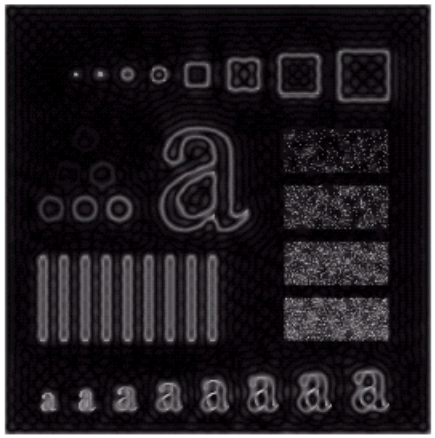
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Ideal High Pass Filters (cont…)

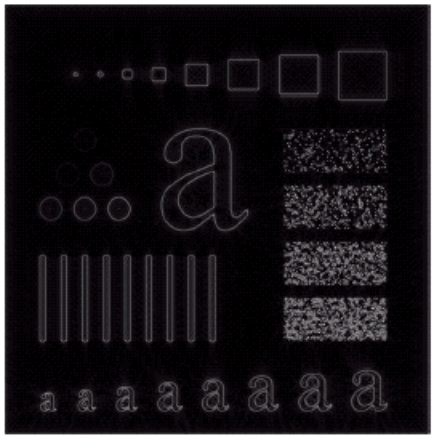




Results of ideal high pass filtering with D_0 = 15



Results of ideal high pass filtering with D_0 = 30



Results of ideal high pass filtering with D_0 = 80

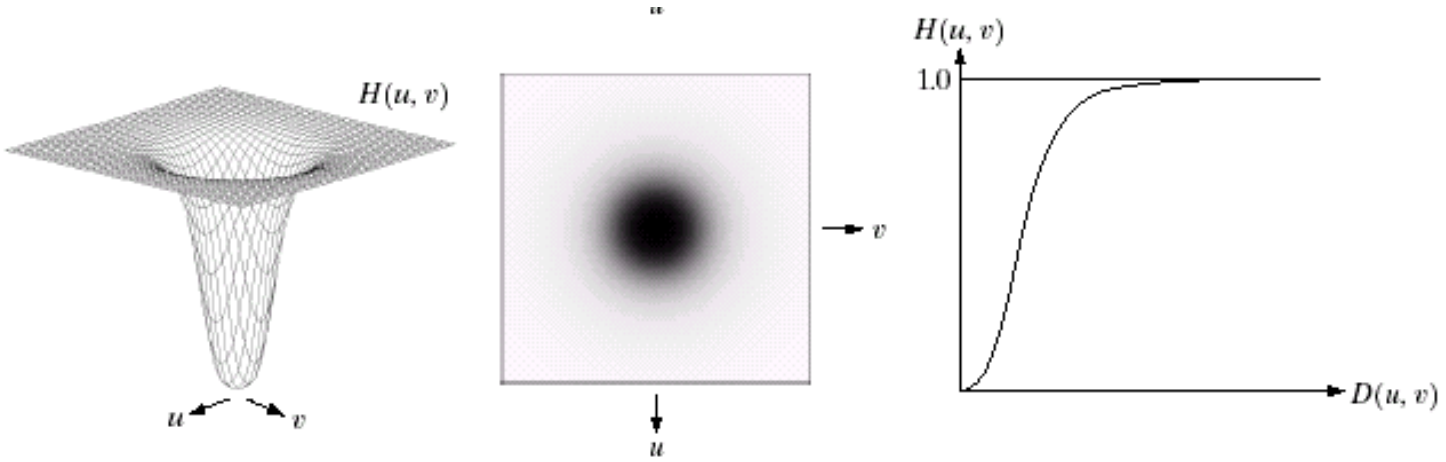
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

4.9.2 Butterworth High Pass Filters

The Butterworth high pass filter is given as:

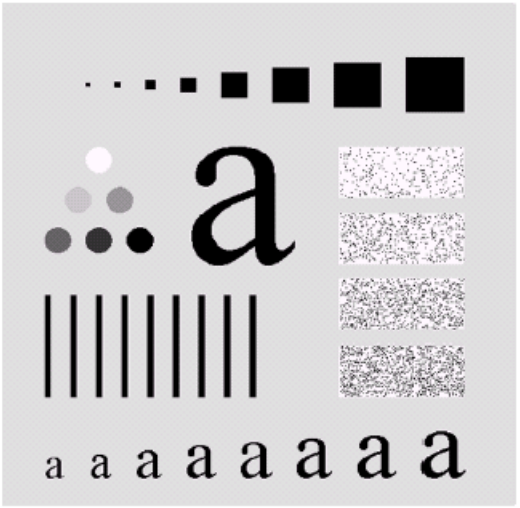
$$H(u,v) = \frac{1}{1+[D_0 / D(u,v)]^{2n}}$$

where n is the order and D_0 is the cut off distance as before

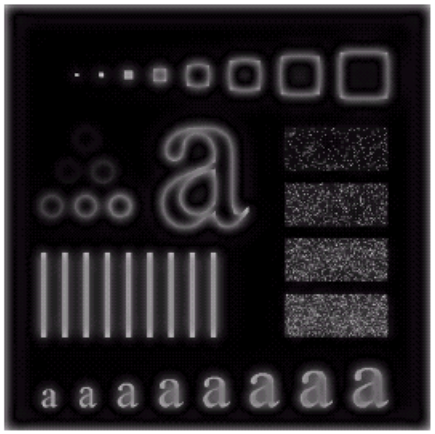


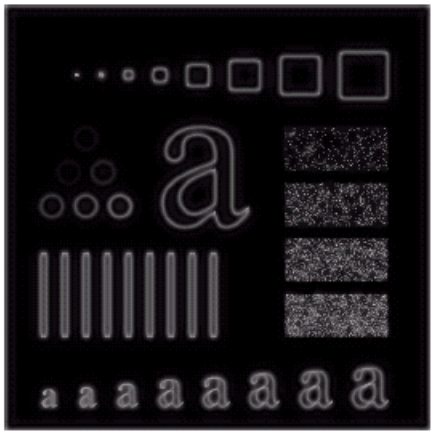
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Butterworth High Pass Filters (cont…)



Results of Butterworth high pass filtering of order 2 with D_0 = 15







Results of Butterworth high pass filtering of order 2 with D_0 = 80

Results of Butterworth high pass filtering of order 2 with D_0 = 30

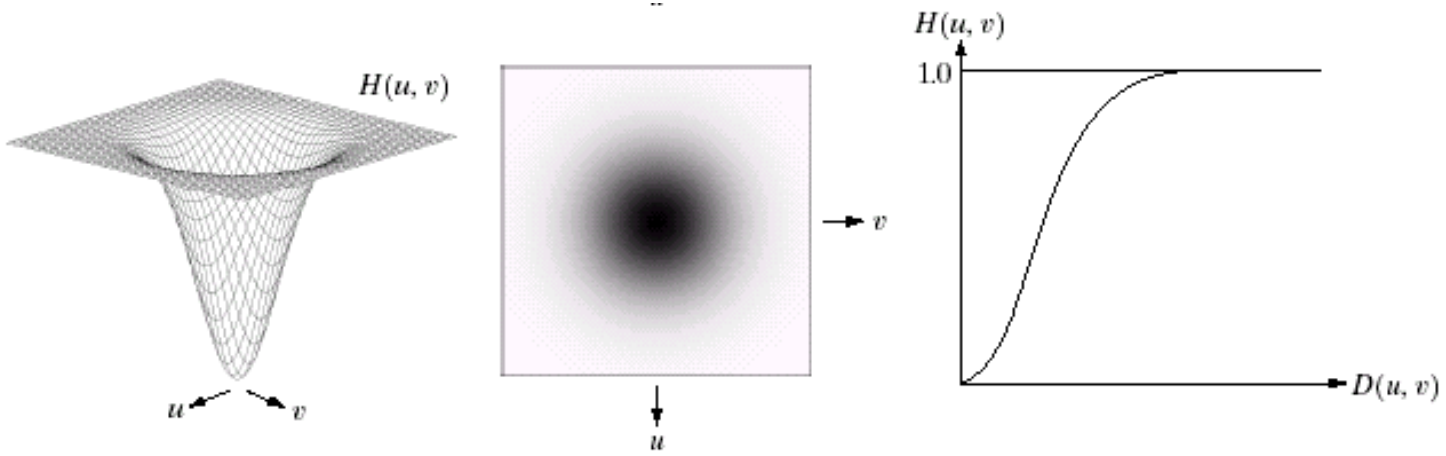
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

4.9.3 Gaussian High Pass Filters

The Gaussian high pass filter is given as:

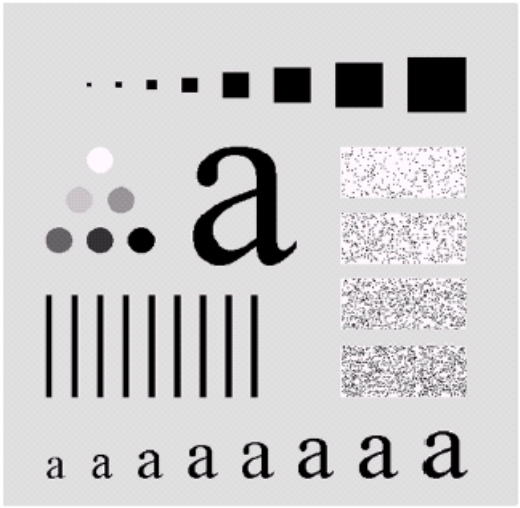
$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0{}^2}$$

where D_0 is the cut off distance as before

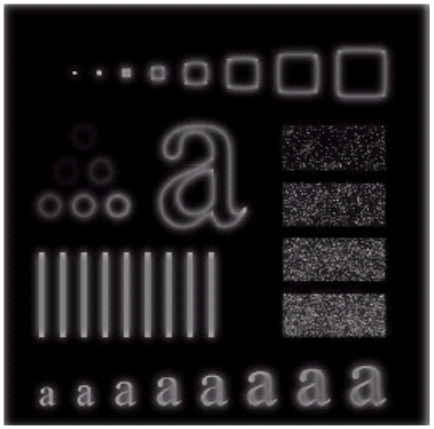


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Gaussian High Pass Filters (cont…)



Results of Gaussian high pass filtering with D_0 = 15





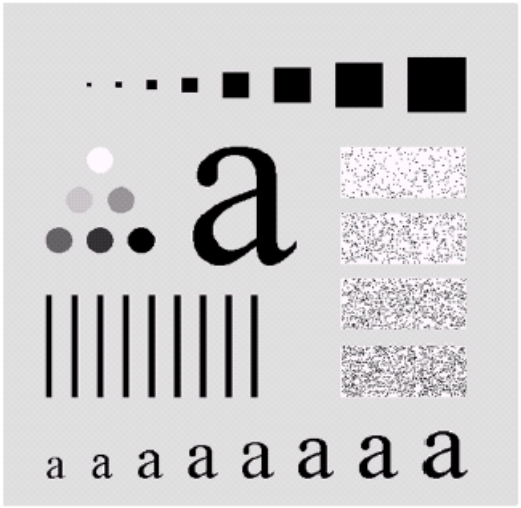


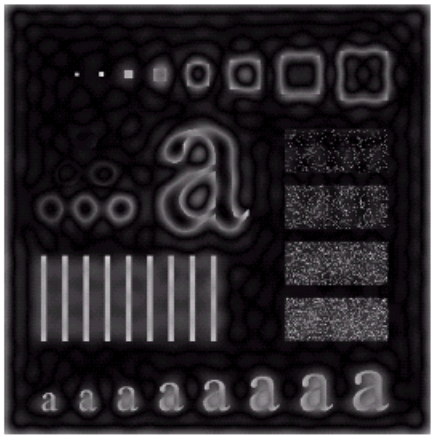
Results of Gaussian high pass filtering with D_0 = 80

Results of Gaussian high pass filtering with D_0 = 30

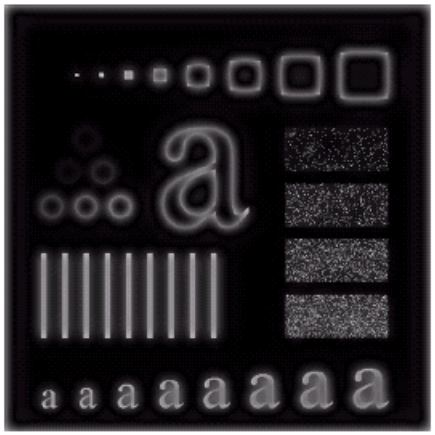
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Highpass Filter Comparison

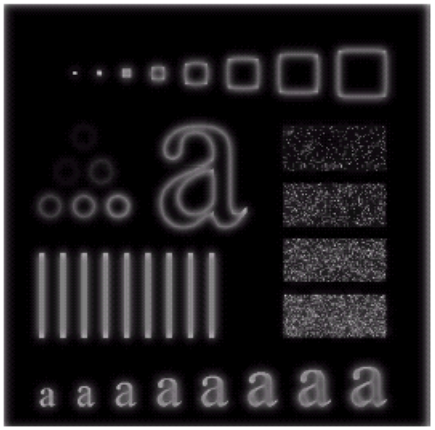




Results of ideal high pass filtering with D_0 = 15



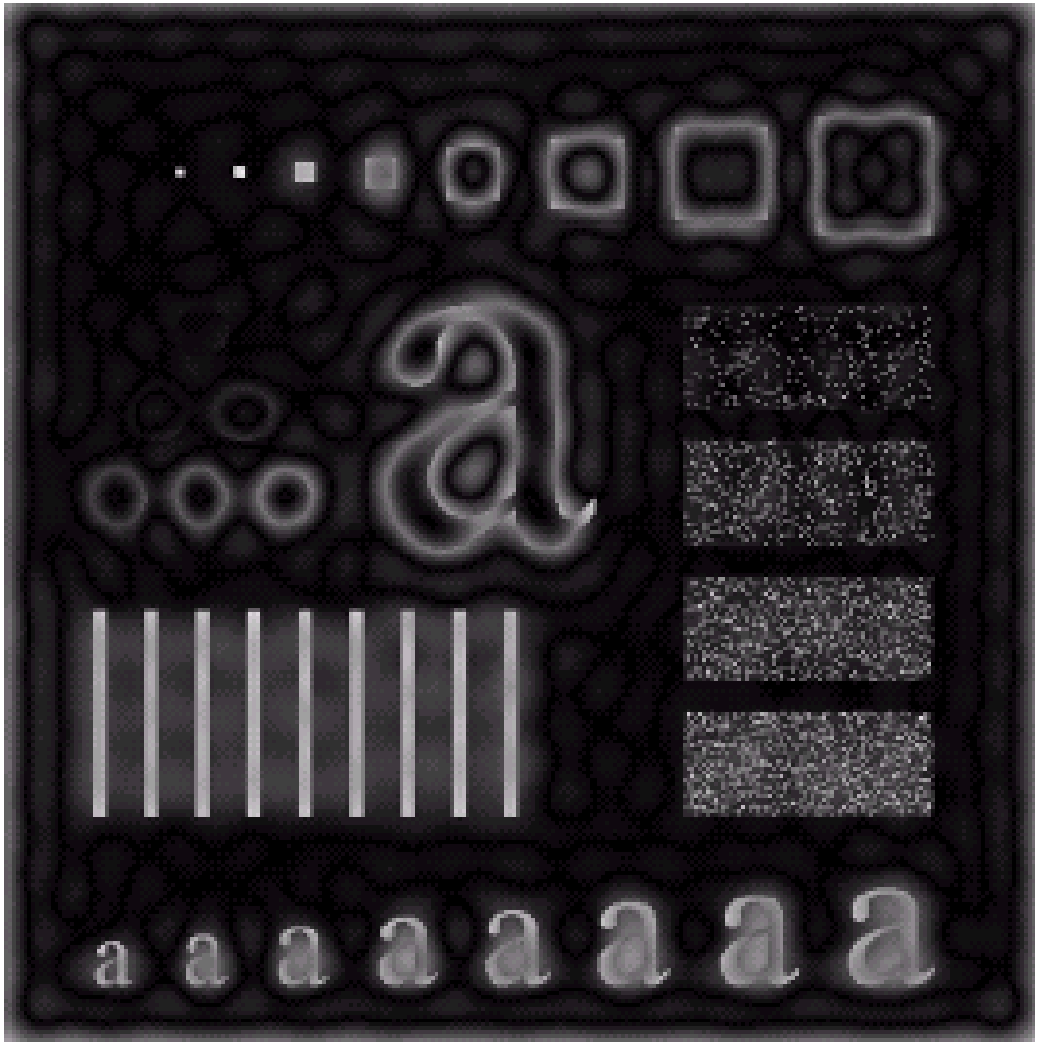
Results of Butterworth high pass filtering of order 2 with D_0 = 15



Results of Gaussian high pass filtering with D_0 = 15

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

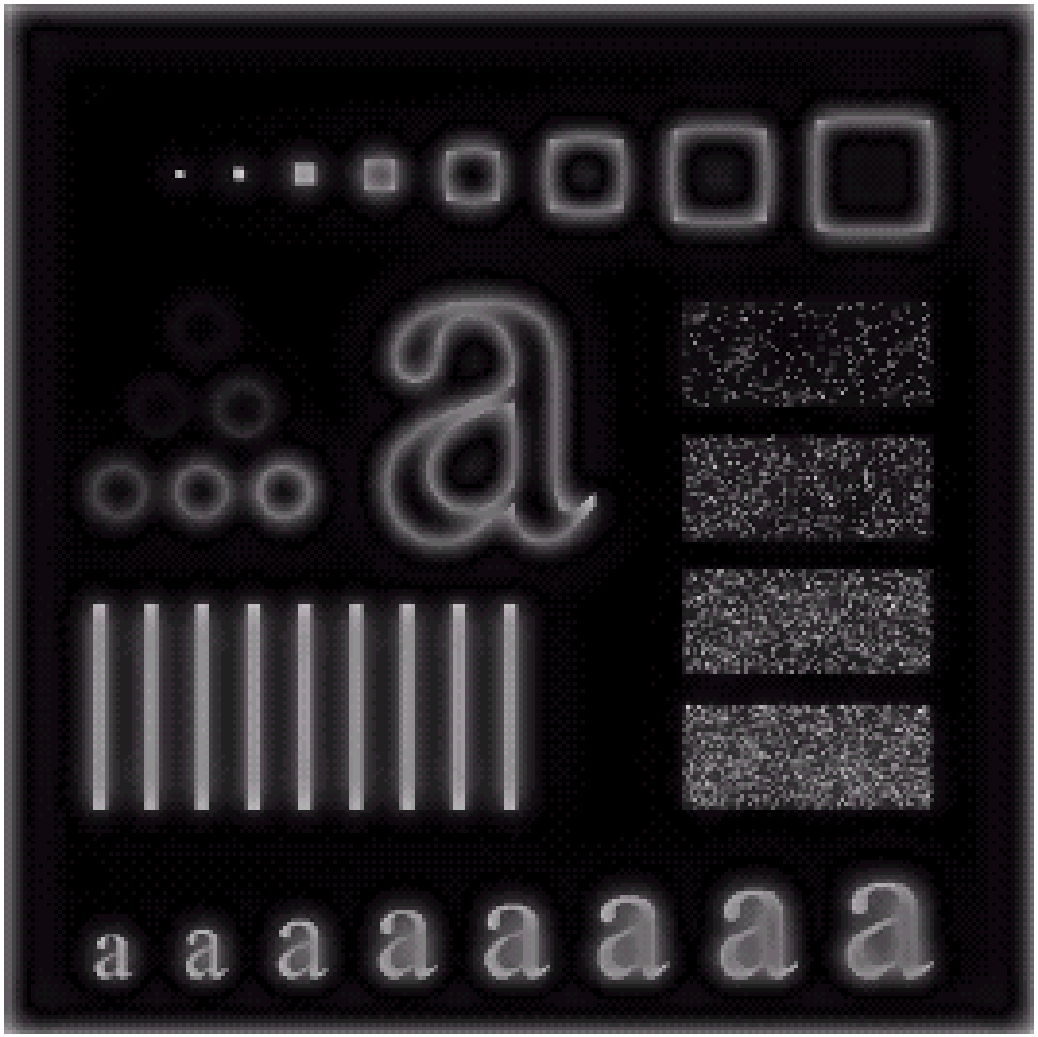
Highpass Filter Comparison



Results of ideal high pass filtering with D_0 = 15

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

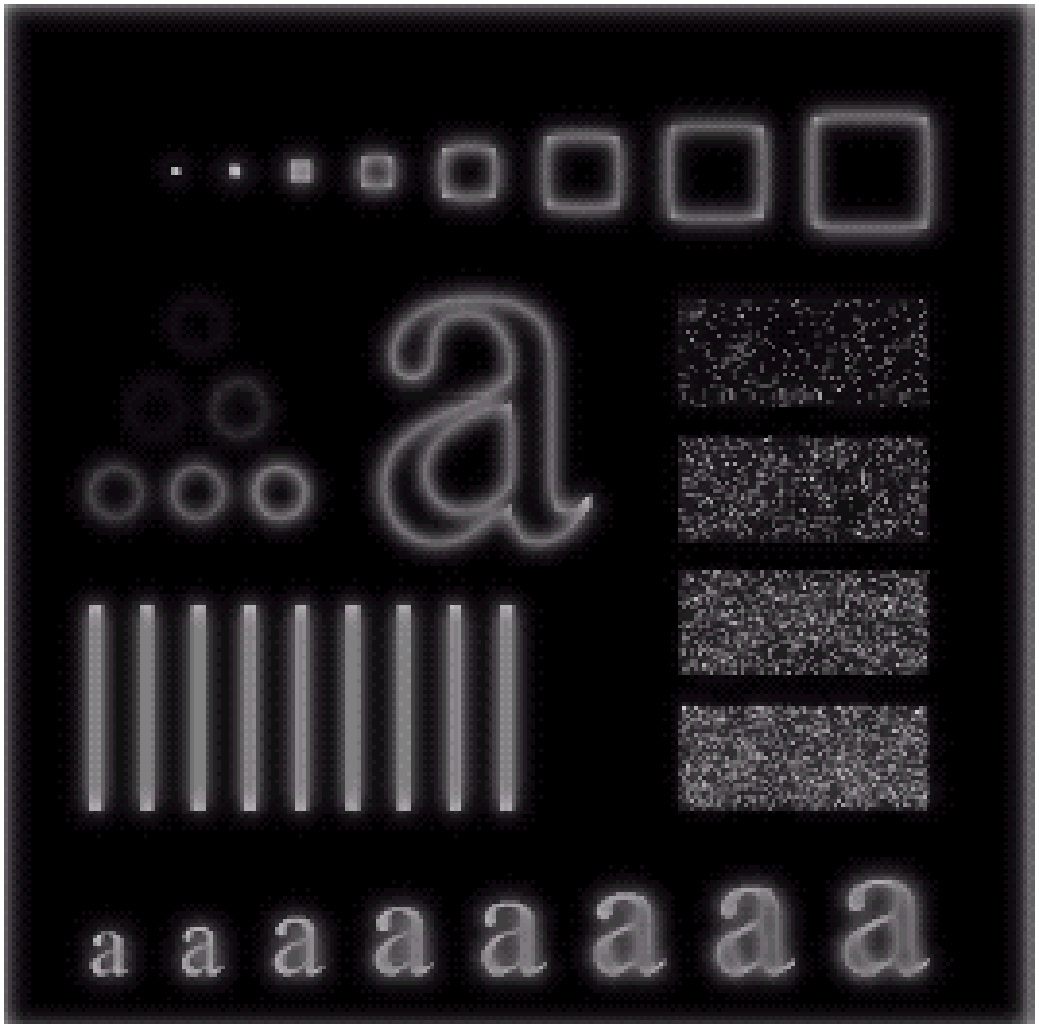
Highpass Filter Comparison



Results of Butterworth high pass filtering of order 2 with D_0 = 15

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Highpass Filter Comparison



Results of Gaussian high pass filtering with D_0 = 15

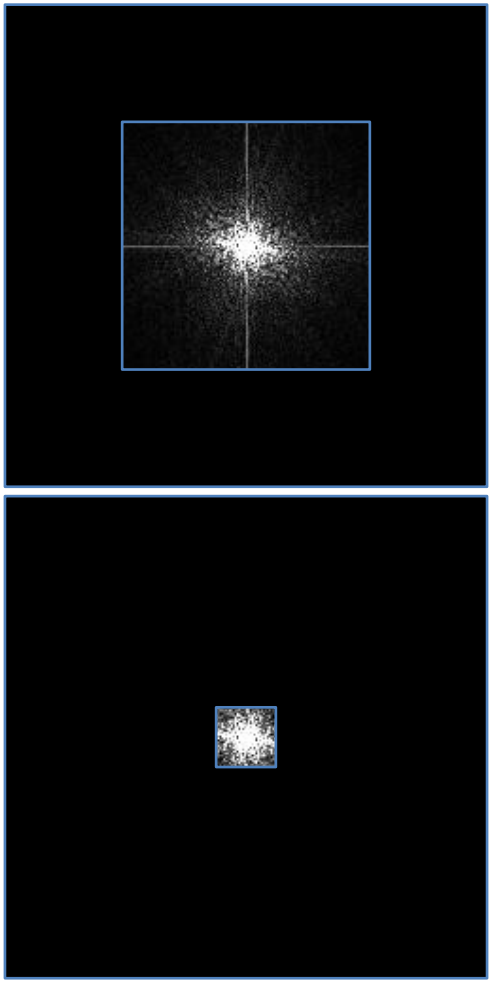
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

DFT在图像处理中的应用

在图像滤波中的应用

- 能量聚集在中间：低频部分
- 低通滤波





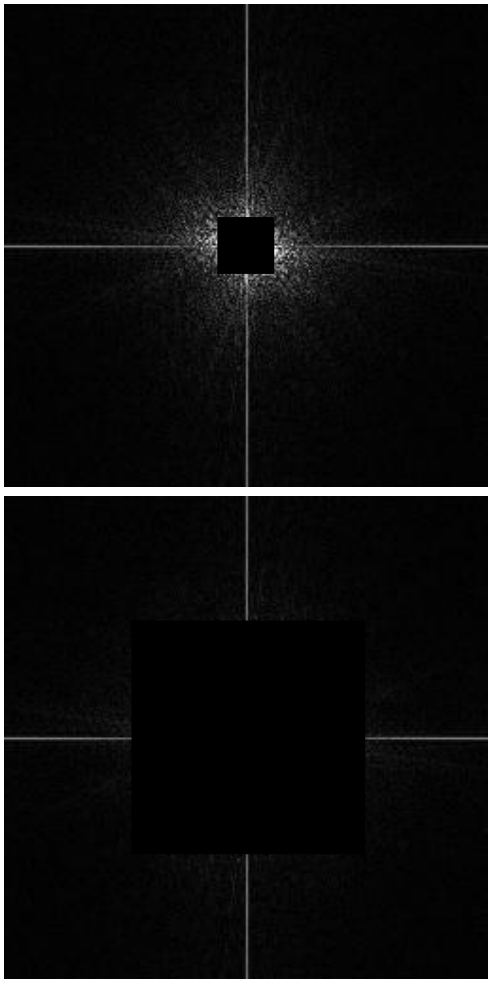


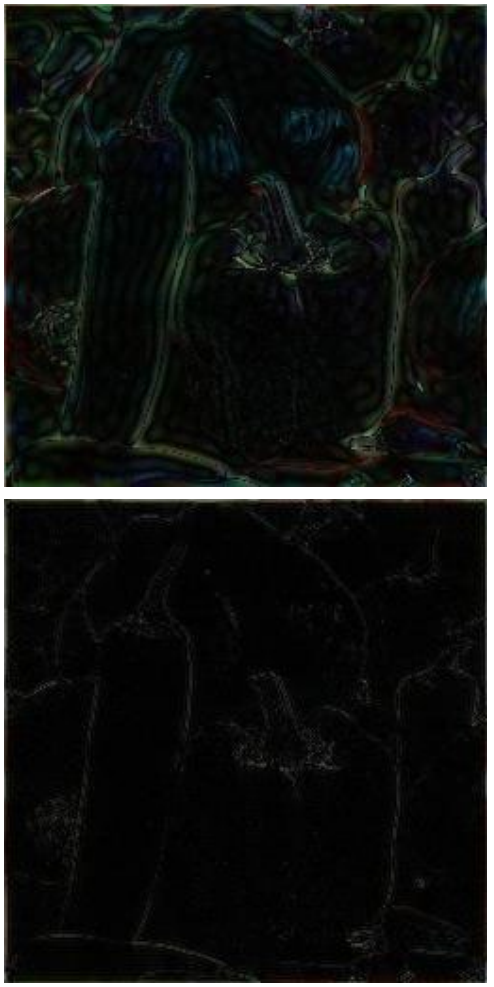
DFT在图像处理中的应用

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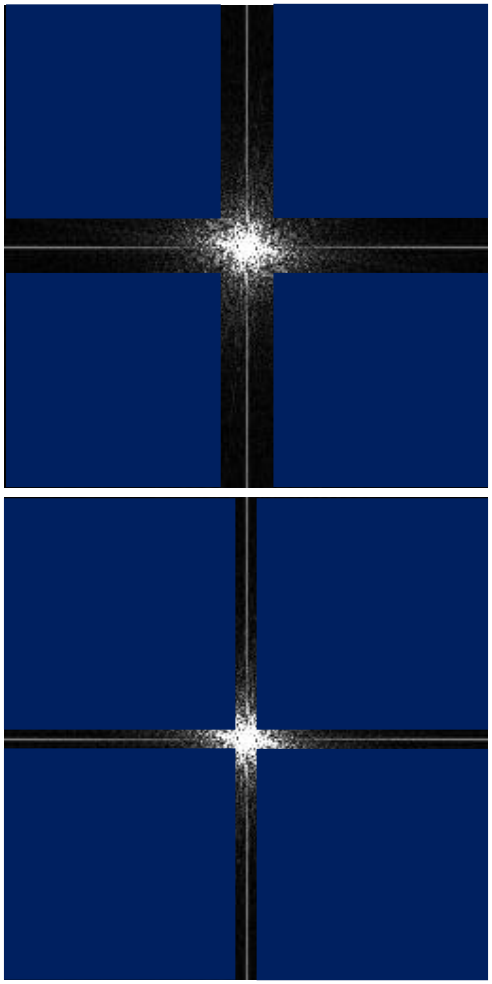


DFT在图像处理中的应用

在图像压缩中的应用

- 能量聚集在中间：低频部分







Homomorphic filtering （同态滤波）

$$f(x,y) = i(x,y)r(x,y)$$

Illumination *i(x,y)*, and reflectance *r(x,y)*, components.

$$z(x,y) = \ln f(x,y) = \ln i(x,y) + \ln r(x,y)$$

$$Z(u,v) = F_i(u,v) + F_r(u,v)$$

$$S(u,v) = H(u,v)Z(u,v)$$
$$= H(u,v)F_i(u,v) + H(u,v)F_r(u,v)$$

filtering

同态滤波

入射光对应图像频域的低频分量；

反射光变化迅速，对应图像频域的高频分量。反映图像细节和局部特性。

滤波器H(u,v)设计思路：

- 抑制低频，压缩图像的动态范围
- 增强高频，加大各部分间对比度
- 中频部分拉伸，增加对比度

Homomorphic filtering

$$s(x,y) = i'(x,y) + r'(x,y)$$

$$g(x,y) = e^{s(x,y)} = e^{i'(x,y)} e^{r'(x,y)}$$

Summary of steps in homomorphic filtering

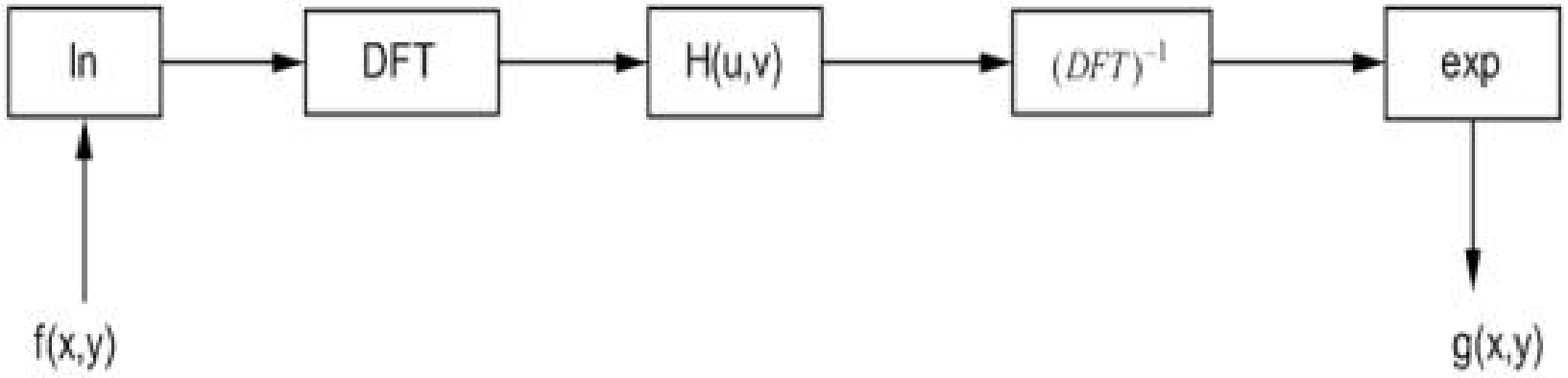


Image enhanced using homomorphic filtering





Image enhanced using homomorphic filtering



